

RUNTIME

LIVE MACHINE PRO

v 1.15.37

*On Air. **In Control.***

USER MANUAL

Second Edition · Version 1.15.37 · English

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REAL-TIME AUDIO SHOW-CONTROL SYSTEM

Runtime Live Machine Pro

v1.15.37

User Manual

Simone Pizzi

Second Edition · 2026

Aligned with version 1.15.37



Runtime Live Machine Pro — User Manual

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CHAPTER 1

Runtime Live Machine Pro: a philosophy

Author's note

Fifteen years of open microphones leave a distinct mark on anyone who has lived through them. I have run podcasts, hosted talk shows, kept a web radio on the air, and for much of that time I did everything alone: the running order, the music, the interviews, the levels, the timing. I know what it feels like to realize, live, that the song is about to end while you are still shaping the thought you want to express. I know what it means to pull down a fader with one hand and find the right clip with the other, while the third hand — the one you don't have — is supposed to hold the thread of what you are saying.

Runtime Live Machine Pro grew out of that frustration, and out of a simple conviction: audio production shouldn't be a job in its own right. It should be transparent. The presenter, the podcaster, the content creator hosting a late-night talk show alone, with no engineer to lean on, needs to concentrate on what they do best: talking, thinking, holding a rapport with the listener. The software takes care of the rest.

I built into RLMP the rules a good sound director applies automatically: the hierarchy between audio events, the ducking that kicks in when you speak, the music that stops and resumes at the right moment. Complex rules, hidden beneath an interface that asks for a single gesture: clicking the right clip at the right time.

This software is designed above all for people running small and mid-sized talk radio, for anyone producing podcasts with professional ambition, for anyone streaming live without a technical crew around them. That said, it isn't exclusive by nature: people working in more structured settings will find tools suited to their needs too. The goal is single-minded: to make the presenter independent of the support roles that aren't always there, and aren't always needed.

Every tool is born from an answer. Runtime Live Machine Pro answers a precise problem: live audio production (radio, podcasts, events, theatre) is a performance activity, not an automation task. It demands instant control, steady nerves and software that won't betray you at the wrong moment.

The software installed on your computer is not a 24/7 music scheduler, nor a DAW for post-production, nor a simple player with a queue. It is a **real-time broadcast machine**, built around the idea that every show is a one-off, unrepeatable act that deserves a dedicated container and precise control over every transition.

1.1 Who it was built for

Runtime Live Machine Pro is aimed at two kinds of users who, despite their different contexts, share the same fundamental need.

The **broadcast professional** — the director of a commercial radio station, the engineer of an audio or video live stream, the presenter running their own show — will find in RLMP a system on a par

with high-end professional tools, with an operational agility those systems often sacrifice on the altar of complexity.

The **content creator** — the independent podcaster, the web-radio host, the live-event organizer — will find a tool that doesn't take years of technical training to master, yet makes no compromises on the quality of the result.

Both get an interface that responds to the key instantly, a stable audio engine, and a save system that doesn't forget.

1.2 The "Single Show" philosophy

The founding concept of Runtime Live Machine Pro is the **isolated project**. Every show you produce — a podcast episode, a live radio broadcast, a theatre performance — lives in a self-contained `.lmp` file that holds everything: the clip layout, the volumes, the MIDI mappings, the cue points, the production notes. When you load that file, you find the show exactly as you left it.

This approach has concrete consequences. You don't have to reconfigure the software every time you move from one show to another. You can carry a project to any computer through the Export project with audio feature and know it will work. You can archive past episodes and reopen them months later without surprises.

The `.lmp` file doesn't contain the physical audio files: it stores their paths on disk. To move a project between computers, the **Export project with audio** feature physically copies everything needed into a self-contained folder.

1.3 The Main-Side-Heavy architecture

Understanding the internal architecture isn't essential to using the software, but it helps explain why certain problems common to other players don't occur here.

Runtime Live Machine Pro is built on **Electron**, a platform that cleanly separates the main process (Main Process, in Node.js) from the interface rendering process (Renderer Process). This separation is used deliberately.

All the heavy operations — audio decoding via FFmpeg, reading files from disk, analysing waveforms, managing backups — are delegated to the Main Process. The Renderer deals solely with the interface: displaying clips, animating the VU meters, responding to clicks. The result is an interface that stays fluid even during intensive operations, and an audio engine that doesn't compete for resources with the pixels on screen.

The custom `media://` protocol keeps audio files from ever being loaded entirely into RAM: they stream straight from disk to the player. You can handle uncompressed WAV files hours long without the application's memory footprint changing appreciably.

1.4 The broadcast grid: a visual grammar

RLMP's operating interface is organized into vertical columns, each with a precise semantic role. Before you even launch the software, it's worth fixing this grammar in mind.

Six columns are visible in the main grid. A seventh surface, the **pad FX** (the effects jingle machine), lives outside the grid, in a dedicated panel described in Chapter 7.

Column	Colour	Function
Show Assets	Green	Idents, beds, structural backing tracks for the show
Jingle	Amber	Recurring identifying jingles and stingers
Promo	Cyan	Promos, self-promotion, scheduled announcements
Episode Songs	Red	The music playlist
Voice / Recordings	Orange	Interviews, voice messages, spoken segments
Pre-Show	Purple	Warm-up music before going live, with jingle and promo rotation

The first three columns (Show Assets, Jingle and Promo) share the same audio nature: they are structural and service elements, treated identically by the mixing engine. The distinction is organizational: keeping idents separate from jingles and promos keeps the running order readable even when it's crowded.

Each column has distinct audio behaviours — mixing priority, exclusion rules, fade values — detailed in Chapter 6. For now it's enough to know that a clip's position in the grid is not decorative: it determines how the software will treat it on air. Columns you don't need can be hidden from view (Settings → General → Broadcast layout) without losing the clips they contain.

1.5 Current version and updates

This manual describes version **1.15.37** of Runtime Live Machine Pro. At startup, the software silently checks whether a newer version is available and, if it finds one, opens an update notice, never during a live show. The update system is described in Chapter 12. The `.lmp` project files are compatible with later versions: updating the software doesn't entail losing or manually migrating existing projects.

CHAPTER 2

Installation and first launch

Installing Runtime Live Machine Pro is designed to require the least possible interaction: a few clicks, no manual configuration, no prerequisites to install separately. The audio engine (FFmpeg) is bundled into the installation package and requires nothing from you.

2.1 System requirements

Before proceeding, check that your computer meets the minimum requirements. The recommended specifications ensure the best experience during long sessions or with many clips loaded at once.

	Minimum	Recommended
Operating system (Windows)	Windows 10 64-bit	Windows 11 64-bit
Operating system (macOS)	macOS 11 Big Sur, building from source	macOS 13 Ventura or later
Operating system (Linux)	Ubuntu 20.04 / Debian 11	Ubuntu 22.04 LTS
RAM	4 GB	8 GB or more
Disk space	300 MB (application)	1 GB + space for audio files
CPU	Any modern dual-core	Quad-core or better

Ready-made packages are available for Windows and Linux. There is no official installer for macOS: the program is built from source (section 2.3).

A dedicated sound card is not required: RLMP works with any audio device recognized by the operating system, from the built-in sound card up to professional USB mixers such as the Rødecaster Pro or the RØDECaster Duo.

2.2 Installation on Windows

1. Download the file `Runtime-Live-Machine-Pro-1.15.37.exe` from the project's **Releases** page on GitHub (`github.com/Pitz72/RRLMP/releases`).
2. Double-click the executable. The NSIS installer starts and copies the files to the appropriate directories.
3. When it finishes, a shortcut is created on the Desktop and in the Start menu.
4. The application launches automatically once installation is complete.

A note on Windows SmartScreen. The installers are not signed with a commercial certificate, so SmartScreen doesn't recognize them automatically. If the warning "Windows protected your PC" appears, click More info and then Run anyway. The software is free of malware; the official installers are published exclusively on the project's Releases page, and the source code is public.

2.3 macOS: building from source

There is no official installer for macOS. Runtime Live Machine Pro is free software: if you have a Mac, you can download the source code and build the program on your own computer.

1. Install **Node.js 20** (and git, if you want to clone the repository).
2. Download the code from `github.com/Pitz72/RRLMP`, with the Code button or with `git clone`.
3. In the project folder run, in order: `npm ci`, `npm run build:main`, `npm run build:preload`, `npx vite build` and `npx electron-builder --mac --publish never`.
4. The `.dmg` file is in the `builds/` folder: open it and drag the application into Applications.

A package built this way is not signed: on first launch macOS shows a Gatekeeper warning. Right-click the application and choose Open, or allow it from System Settings → Privacy & Security, in the Security section. Always up-to-date instructions are in the project's `CONTRIBUTING.md` file.

> **NOTE**
> An application you build yourself reports new versions but can't install them: to update, download the updated code and build again (Chapter 12). `.lmp` projects stay compatible.

2.4 Installation on Linux

Two distribution formats are available:

- **AppImage** — a portable executable, no installation required. Make the file executable (`chmod +x`) and launch it directly.
- **.deb package** — for Debian/Ubuntu/Mint distributions. Install with `sudo dpkg -i filename.deb` or open it with the graphical package manager.

On some distributions you may need to install the `libasound2` package for ALSA audio support. Consult your distribution's documentation if the application won't start.

2.5 The welcome screen

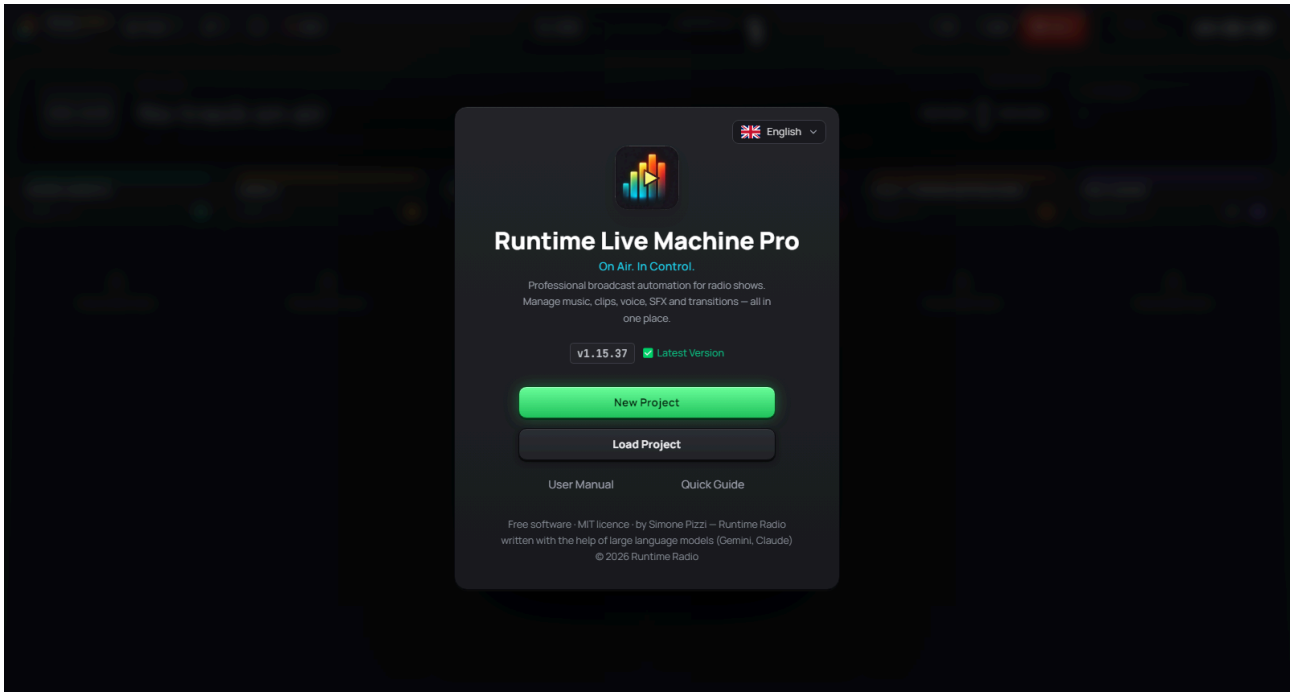


Figure 2.1 — The welcome screen: software identity, update status, main actions and the language drop-down in the top-right corner.

On first launch — and at every subsequent launch, until you open a project — RLMP presents the **welcome screen**, the gateway to all preliminary operations. In the centre you find the software identity and actions; in the top-right corner, the language drop-down.

Identity and actions. The software logo (the bars of a VU meter with the play symbol) identifies the Pro edition. Below the title and slogan is the installed version number, accompanied by the status of the update system:

- **"Latest Version"** (green, with a tick) — you are running the most recent version available.
- **"Update Available v..."** (amber, bouncing) — a newer version exists, shown by its number. This is a button: click it to open the update window (Chapter 12).
- **"OFFLINE"** (light red) — the update service could not be reached; the software works all the same.

Below that are the main actions:

- New Project — creates an empty session with the columns ready to load.
- Load Project — opens an existing `.lmp` file. Before making it operational, RLMP runs an **integrity check**: it verifies that every referenced audio file still exists at the stored path. Clips whose file is missing are flagged straight away: the card turns dark red, with the MISSING FILE badge.
- User Manual — opens this manual as a PDF in the browser, in the interface language (English or Italian). An internet connection is required.
- Quick Guide — a short getting-started guide that opens inside the software and can be read offline too.

Language (top-right corner). RLMP supports two interface languages: English and Italian. The drop-down in the top-right corner shows the flag and name of the active language: click it and pick the other one. The selection takes effect immediately and is remembered from one session to the next.

2.6 The first launch: what to expect

When you first open a project, you'll find the logo at the top left, with the **RLM PRO** wordmark and the version number. Behind the interface, opening the project starts the audio engine in the background: FFmpeg is initialized and the `media://` streaming protocol begins listening, ready to serve files from disk without loading them into memory.

The software starts preferably in full-screen mode. If the window opens resized, press `F11` (Windows/Linux) or `Ctrl+Cmd+F` (macOS) to bring it full screen — the ideal condition for production work.

The **On Air timer** in the header stays at `--:--:--` until the first clip of the session is launched. From that moment it starts counting the time elapsed on air: a useful reference for anyone working with fixed-time running orders.

CHAPTER 3

The working interface

The Runtime Live Machine Pro interface is designed like a hardware broadcast surface: raised keys, recessed displays, LEDs in the colour of each column. At rest it stays quiet, and whatever goes on air lights up. The dark theme, the high contrast and the large type are there so you can read everything at a glance, even in the busiest moments of a live show.

When you open a project, the screen divides into three bands: the **Control Bar** at the top, which manages the project and the system; the **ON AIR display** just below it, which puts the playing track front and centre; and the **broadcast grid** in the centre, where the actual work happens.

3.1 The Control Bar (Header)

The header spans the full width of the screen and is split into three zones:

- **on the left**, project and archive: the logo, the **FILE** menu, the **Tools** menu, the **LOG** and the **REC** recording button;
- **in the centre**, exactly halfway across the screen, the audio console: microphone (**ARM**), VU meter and master volume;
- **on the right**, live controls and time: **FX**, **MIX**, **STOP ALL**, the **On Air** timer and the clock.

On screens at least 1800 pixels wide (Full HD, for example) the extra space goes to readability: the VU LEDs are larger, the master fader is longer and therefore more precise, timer and clock are larger, the buttons taller, and the log also shows the LOG label next to its icon. On narrower screens the bar stays compact and still fits at 1366 pixels.

Identity

Logo and version. At the far left, the logo sits beside the wordmark **RLM PRO**, with “PRO” in amber. Below it, in small type, is the installed version (`v1.15.37`). Hover over the logo to reveal the full software name and version number.

File menu

The **FILE** button opens a menu with the project operations:

- New Project — opens an empty session. If there are unsaved changes, the software asks for confirmation.
- Save Project — a quick save to the current `.lmp` file.
- Save As… — always opens the dialog box, for creating progressive versions (e.g. `Ep47_draft.lmp`, `Ep47_final.lmp`).
- Export project with audio — consolidates all the audio inside the project (an `audio/` subfolder) and repoints the clips to it, so you can safely delete the originals. Described in Chapter 10.
- Load Project — opens a `.lmp` project from disk.
- Import M3U — imports a playlist in M3U format as a sequence of clips.

When there are unsaved changes, an amber dot lights up on the corner of the FILE button and, in the menu, the Save Project entry turns amber. After an automatic save, the green Auto-saved label appears briefly below the button.

Tools

Next to FILE, the button with the **wrench** icon opens the Tools menu:

- Undo and Redo — the running-order edit history (`Ctrl+Z` / `Ctrl+Y`).
- MIDI Learn — enables MIDI learning mode (Chapter 8). To the right of the entry you can see how many MIDI controllers are connected.
- Keybinds & MIDI Dashboard — the window for assigning keys to clips.
- General Settings — the software's global preferences (Chapter 13).
- Info & Updates — version, credits and manual update check.

While MIDI Learn is active, the Tools button keeps showing a label, even with the menu closed: the word MIDI, or the seconds left in the learning countdown.

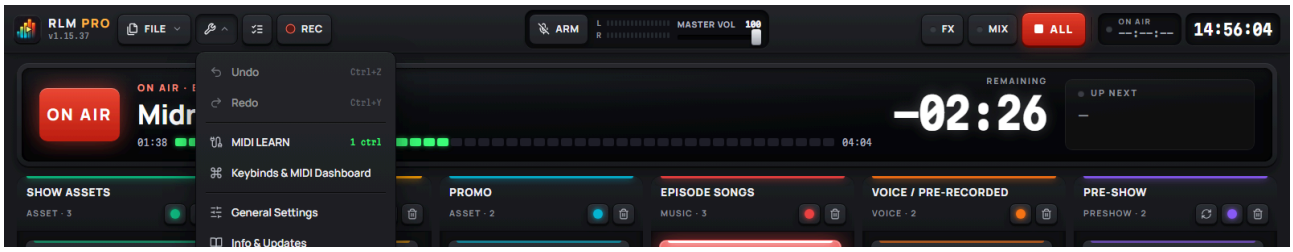


Figure 3.1 — The Control Bar with the Tools menu open: logo, FILE, Tools, LOG and REC on the left; the audio console in the centre; FX, MIX, STOP ALL, On Air and clock on the right.

Log and recording

LOG. Opens the **Playout Log**, the chronological record of the tracks that went on air (Chapter 13). You consult it before or after the show, which is why it sits with FILE and Tools.

REC. Starts and stops the session recording (Chapter 9). You press it once, at the start of the show. While recording, the button turns red and shows the running time; once recording stops it gives way to the export controls.

Microphone, levels and transport

ARM (microphone). Arms and disarms the microphone for **Smart Mic** (Chapter 6). Disarmed, the button shows a crossed-out microphone icon. Armed, the icon is replaced by a small eight-LED VU with the microphone level, and the **Mic Vol** slider appears next to it; while you speak, the button turns red and pulses. The **On Mix** label, on the bottom edge of the button, shows that the voice is also going into the mix. Smart Mic must be enabled in the Microphone tab of the Settings (Chapter 13): if you turn it off, the microphone is disarmed.

Stereo VU meter (L/R). Two rows of 16 LEDs, one per channel, show the real output level after the Master Volume. Each LED is worth 3 dB: the LEDs are green up to −12 dBFS, yellow up to −3 dBFS, and the last one is red, right below the limiter. The peak LED stays lit for about a second and a half, so you can see even the shortest peaks. If the last LED stays lit for long, the signal is pushing against the limit: lower the level.

Master Volume. The **MASTER VOL** fader controls the software's overall output volume, from 0 to 100; the value is shown next to the label. It acts as a master fader: brought to zero, no sound comes out, regardless of the state of the individual clips. If you have mapped a MIDI control to the Master Volume, the assignment (for example `C7`) appears in place of the value.

FX. Opens and closes the pad FX, the effects jingle machine (Chapter 7). The LED on the button is green while the pad is open; a small counter shows how many effects are playing at that moment.

MIX. Opens and closes the Automix view, the deck dedicated to the Music column (Chapter 7). Here too, the green LED means the view is open.

STOP ALL (red "ALL" button). Instantly stops every active clip and cancels any fades in progress. It is the system's emergency command. The `Esc` key does the same when the application is in focus, even while you are typing in a text field.

> **NOTE**
> Unlike previous versions, `Esc` is no longer registered as a system-wide global shortcut: it acts when RLMP is the active window. This choice lets dialog boxes use `Esc` to close without stopping the live show.

Session indicators

At the far right you find the **On Air timer** and the **studio clock**.

The On Air timer is a small display labelled ON AIR: at rest it shows `--:--:--`; when the show starts, a red LED pulses and the time runs in `HH:MM:SS` format. The studio clock is in 24-hour format, synchronized with the system clock.

Unobtrusive notifications (**toasts**) about completed operations or system warnings appear at the bottom right. Unlike blocking dialogs, toasts disappear on their own after a few seconds and don't interrupt playback.

3.2 The ON AIR display

Between the Control Bar and the grid sits a full-width display that shows the track on air, readable even from across the room. It is **always there** and has a fixed height, so the columns below it never move. With nothing playing it shows **OFF AIR**, the words "No track on air" and an unlit lamp.

From left to right you find:

- **ON AIR** — the lamp: red and lit while a track is on air, grey and unlit at rest.
- **Title and column** — the "ON AIR" label followed by the column name, the clip title in large type and, next to it, the artist (or "In loop" for looping clips).
- **LED bar** — below the title, a row of LEDs that light up as the track plays, with the elapsed time on the left and the total duration on the right. If the clip has Intro and Outro markers (Chapter 5), the LEDs before the end of the intro are cyan and those from the Outro on are amber; two vertical ticks, one cyan and one amber, mark the two points. The coloured zones are visible even while the LEDs are off, so you know in advance where the Outro falls.
- **REMAINING timer** — the time left until the end of the track, in a colour that changes with the phase: **cyan** during the intro (counting down to the end of the intro), **white** in the body of the track, **amber** from the Outro point on, **flashing red** in the last 10 seconds. For looping clips it shows "LOOP". In the 15 seconds before the Outro, the amber **OUTRO IN** label appears below the timer with its countdown.
- **UP NEXT** — the clip that will start next, with artist and duration, when the clip on air is set to Play Next: the LED next to the label turns green. Otherwise the box stays unlit and shows a dash.

If several clips are playing together, the display shows the one from the highest-priority column, in this order: Episode Songs, Pre-Show, Voice, then the other columns. Pad FX effects never appear on the display. When one track hands over to the next in a sequence, the display holds the track that just ended for a moment instead of flashing OFF AIR.

3.3 The six-column grid

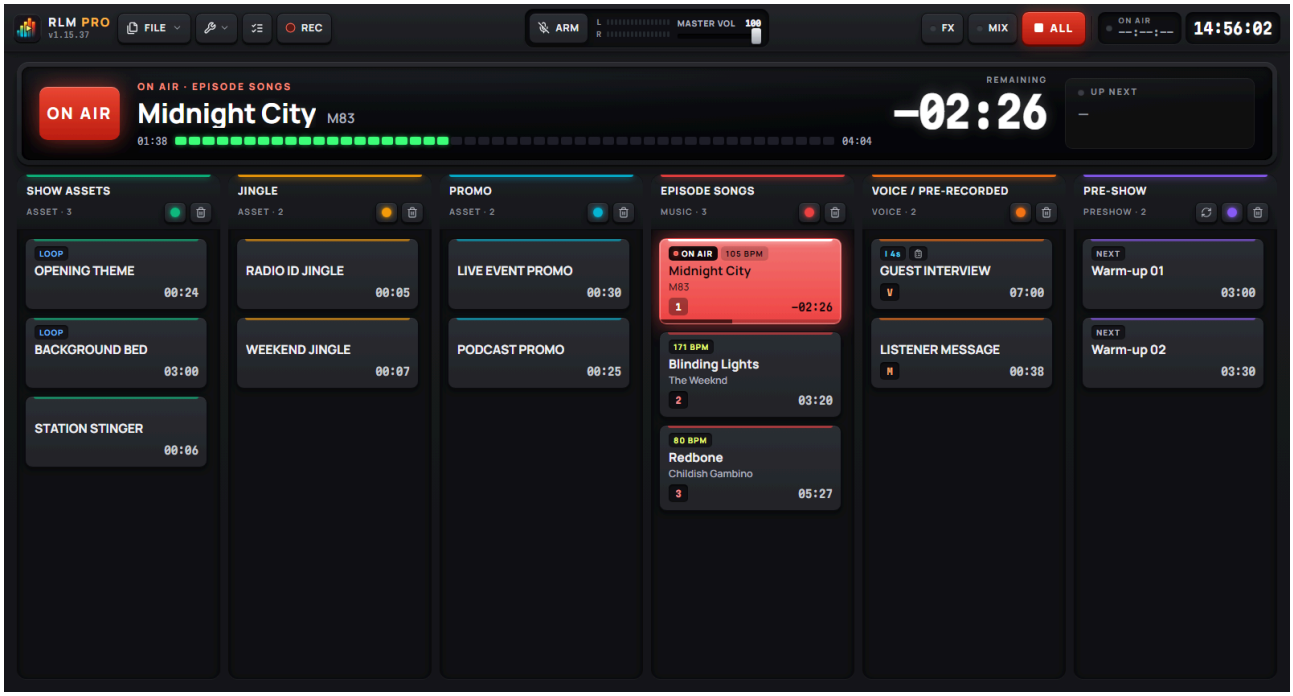


Figure 3.2 — The working interface: the ON AIR display with a track on air and the six-column grid. The clip on air is lit in the colour of its column.

The grid is the software's operational centre: six vertical columns side by side, each with its own header, its own colour and its own audio-behaviour logic. Sound effects have no column in the grid: they live in the pad FX (Chapter 7).

Column headers

The header has two rows, with a glowing strip in the column's colour along its top edge. The top row holds the full column title. The bottom row shows the type (ASSET, MUSIC, VOICE or PRESHOW) with the number of clips and, on the right, the column's buttons.

The header also doubles as a status indicator. When the playing clip is the last one available in the column, is not looping, and less than **20 seconds** remain until the end, the **DEAD AIR** alarm goes off: the header turns amber, the strip flashes, a warning triangle appears next to the title and the **END** badge appears below it. During the alarm the column's buttons are hidden. It's the advance notice that gives you time to prepare the next track before silence.

Each column's colour is customizable: click the button with the coloured dot, in the second row of the header, to open a palette of **30 shades**. The choice is saved in the project file.

When a column contains at least one clip, a button with a **trash** icon appears in its header: the **Clear column** command removes every clip in that column in one go. For safety it always asks for confirmation, stating how many clips will be removed, and the operation can be reversed with Undo (`Ctrl+Z`). On empty columns the button does not appear.

The **Pre-Show** column header also carries a **rotation** button: when active, the pre-broadcast queue automatically inserts jingles and promos at regular intervals (Chapter 13).

The six columns

Show Assets (Green) The structural elements of the show: idents, backing tracks, beds, institutional stingers. They behave as background elements: they yield space when voices or songs come in, but keep their internal rotation until they are stopped.

Jingle (Amber) and **Promo (Cyan)** Two columns dedicated, respectively, to identifying jingles and to promos or self-promotion. In audio terms they behave exactly like Show Assets (they belong to the same family), but keeping them separate keeps the running order tidy and readable.

Episode Songs (Red) The music playlist. Clips in this column take an active part in the automatic mixing: they are lowered when voices play and, in turn, silence the Asset beds when they start playing (Chapter 6). On music clips the software automatically detects the **BPM**, shown with a dedicated badge.

Voice / Recordings (Orange) Interviews, pre-recorded spoken segments, voice messages. This column has the **highest priority** in the mixing system: when a clip here is playing, all other signals are lowered to a background level.

Pre-Show (Purple) The pre-broadcast warm-up playlist. It works as a self-contained music queue, with optional rotation of jingles and promos. When the show proper begins, this column is typically emptied or disabled.

3.4 The audio card (Clip)

Every imported audio file becomes a **card** in the grid: a raised rectangular key. The card is the operating unit of the system: you see it, launch it, configure it, move it.

Anatomy of a card

From top to bottom, a card shows: a thin strip in the column's colour, the badge row, the title on a line of its own, the artist (only for music clips that have one) and, at the bottom, the assigned keys on the left and the timer on the right. The column type is not repeated on the card: the header already says it.

Title and artist. The file name, or the custom name assigned in the properties. The custom title changes only the label inside the software; the original file on disk stays untouched. For music clips, the artist name may appear below the title.

Timer. At rest, it shows the clip's total duration in `MM:SS` format. During playback it switches to a **countdown**, with the negative prefix (e.g. `-01:20`). When less than 15 seconds remain, the digits turn white on a dark label and flash.

Status badges. Small labels communicate the configured properties at a glance:

- **ON AIR** — the clip is playing: a dark label with a red LED.
- **LOOP** — the clip will restart from the beginning when playback ends.
- **NEXT** — when this clip ends, the next one in the column will start automatically.
- **▶ UP NEXT** — highlights which clip will be next to start in the automatic sequence: the badge is green and flashes, the card has a green ring and its top strip turns green.
- **### BPM** — the detected tempo, on music clips, written in light yellow.
- **I ##s** — the clip has an Intro point configured (Chapter 5): the badge, in cyan, shows its duration in seconds and is always visible, even when the clip is stopped.
- **TRIM…** — silence analysis in progress (Auto-Trim), with an amber LED.
- **FADE OUT** — appears on the outgoing clip during a crossfade or a fade.
- **Clipboard icon** — the clip has a note attached in the NoteBoard (Chapter 13).

Assignments. If the clip has a keyboard key assigned, the key appears at the bottom left, in a dark box with the character in the column's colour; if it has a MIDI binding, the label `M` followed by the note number (e.g. `M60`) appears in cyan.

Structure cues. If markers are configured, during playback the assigned keys give way to the countdowns `INTRO -MM:SS` (cyan label) and `OUTRO IN -MM:SS` (amber label); once the tail has begun, the label becomes `OUTRO`.

The card on air. When a clip goes on air, the whole card lights up in the colour of its column. The text is dark on light colours and white on dark ones, chosen by contrast: a midnight-blue column reads as well as a yellow one. The top strip turns white and a progress bar runs along the bottom of the card.

Interacting with the cards

- **Left click** — starts the clip if it is stopped; stops it (with a fade out) if it is playing.
- **Ctrl + Click** (Windows/Linux) or **Cmd + Click** (macOS) — selects the clip without starting it. A blue ring appears around the card. Useful for multiple selection and bulk deletion.
- **Delete key** (or Delete / Backspace) — removes the selected clips from the grid. If more than one clip is selected, the software asks for confirmation.
- **Right click** — opens the **Clip Settings**: properties, waveform editor, notes (Chapter 5).
- **Drag & Drop** — drag a card to reorder it within the column or move it to another. A glowing blue line shows the insertion point while dragging.

Cards in an error state

A dark red card, with the red **MISSING FILE** badge and the title in light red, signals that the referenced audio file is no longer reachable: it has been moved, renamed, or is on an external disk that is not connected. The clip is not playable until the file returns to its original path. Handling path errors is covered in Chapter 14.

CHAPTER 4

The basic workflow: load and play

The fundamental operating cycle of Runtime Live Machine Pro breaks down into three phases: importing the audio files, organizing them in the grid, playing them during the live show. This chapter describes each phase with the precision needed to work safely even under pressure.

4.1 Importing the audio files

RLMP has no internal browser and no centralized library. Importing is done by **drag & drop** directly from the operating system's file manager (File Explorer on Windows, Finder on macOS, Nautilus or equivalents on Linux). Alternatively, from the FILE menu you can import an **M3U** playlist and turn it into a sequence of clips.

The basic gesture

1. Open the folder on your computer where the audio files are.
2. Select one or more files. To select several files: `Ctrl+Click` for a discontinuous selection, `Shift+Click` for a continuous one.
3. Drag the selected files over one of the grid columns and drop them. For sound effects, drag them straight onto the pad FX (Chapter 7).

Each file creates a card in the destination column. If you drag several files at once, the cards are created in the order the files appear in the file manager, from top to bottom.

Insertion indicator. During the drag, a luminous blue line runs along the column, showing exactly where the cards will be inserted. You can add new clips at the top, at the bottom, or at an intermediate position with precision.

Supported formats

The bundled FFmpeg engine ensures compatibility with a wide range of audio formats:

Format	Extension	Notes
MP3	`.mp3`	All bitrates
WAV	`.wav`	Uncompressed PCM, any bit depth
FLAC	`.flac`	Lossless, any sample rate
AAC / M4A	`.aac` , `.m4a`	Includes files from iTunes/Apple Music
OGG Vorbis	`.ogg`	
Opus	`.opus`	
WMA	`.wma`	Windows Media Audio
WebM / MP4	`.webm` , `.mp4`	Audio tracks held in these containers

A note on performance. The `media://` streaming protocol ensures that audio files are not loaded into RAM at import time. An uncompressed 2 GB WAV file behaves exactly like a 5 MB MP3: loading is instant and the impact on system memory is negligible. CPU resources are engaged only during active decoding, that is, during playback.

The file path

RLMP stores the **absolute path** of the file on disk, not a copy of the file itself. If you move, rename or delete the original file, the corresponding card turns red and is no longer playable. To work

across several computers or create portable archives, use the **Export project with audio** feature described in Chapter 10.

4.2 Playback: starting and stopping clips

Starting a clip

A **left click** on the card is enough to start playback. The feedback is immediate: the card lights up in the colour of its column, the timer switches to a countdown, and the VU meters in the header reflect the output signal.

If a keyboard key has been assigned to the clip (see Chapter 8), that key works as an alternative to the click, handy when you are operating on another part of the interface and don't want to move the mouse.

Stopping a clip

Click the active clip — the clip enters the **fade out** phase and stops within the time set in its properties (see Chapter 5).

`Esc` **key** — stops all active clips instantly. It is the emergency command. It works when RLMP is the active window, even while you are typing in a text field.

STOP ALL button in the header — identical to `Esc`, accessible with the mouse.

The per-column exclusion logic

In most columns, RLMP applies the **"one clip at a time"** rule: if you are playing Track A in the Songs column and click Track B in the same column, Track A stops (with a fade out) and Track B starts. There's no need to manually stop the current clip before starting another. With the MIX view open, in the Music column the change becomes a mix instead, beat-matched when the rhythm of the two tracks allows it (Chapter 7).

The **pad FX effects** are the exception: they overlap everything, including other effects, and don't interrupt whatever is playing. A round of applause can start while a song is playing without cutting off its playback.

4.3 Organizing the running order

Reordering clips

While preparing the show, or even while it's under way, you can rearrange the order of the clips at any time.

Internal dragging. Click a card, hold and drag it up or down within the same column. The blue guide line shows the insertion point. The clip slots into its new position without interrupting anything currently playing.

Moving between columns. You can drag a clip from one column to another. When you do, the clip **inherits the rules of the destination column**: a pre-recorded voice moved into the Songs column will start being ducked exactly like a music track.

Moving clips between columns changes their behaviour, so use it deliberately, especially during a live show.

Multiple selection and deletion

To remove several clips from the grid in a single operation:

1. `Ctrl+Click` (Windows/Linux) or `Cmd+Click` (macOS) on each clip to select. A blue ring appears around the card.
2. Press `Delete` or `Backspace`. The software asks for confirmation if more than one clip is selected.

Deleting from the grid removes the clips from the current project, not the audio files from disk. If you make a mistake, `Ctrl+Z` undoes the operation.

> **TIP**
> Once the show is live, emptying the Pre-Show column with a multiple selection and `Delete` is the quickest way to free up visual space in the interface and switch to operating mode.

4.4 Structure cues: INTRO and OUTRO

Every clip can have two **structural markers** configured in the waveform editor (Chapter 5):

- **Intro Marker** — the point where the track's main melody actually enters, after the instrumental intro. Useful for knowing exactly when to start talking over the intro.
- **Outro Marker** — the point where the track's final tail begins. It flags the right moment to prepare the transition to the next track.

When a clip's playback approaches these points, a visual warning appears at the bottom left of the card:

- **INTRO −MM:SS** — for the whole intro, the time left until the Intro Marker, on a cyan label.
- **OUTRO IN −MM:SS** — in the last 15 seconds before the Outro Marker, on an amber label. Once the tail has begun, the label becomes **OUTRO**.

These warnings appear only if the markers have been configured. On clips without markers, the card shows only the standard countdown to the end of the track.

CHAPTER 5

Clip properties and the Waveform Editor

Every audio file has a history before it reaches the grid: recordings with seconds of silence at the start, tracks with endless tails, interviews with a level too low against the rest of the show. Rather than reaching for an external audio editor every time a file isn't "broadcast-ready", RLMP provides a configuration panel for each clip and a visual waveform editor with trimming and marking features.

Every change made through these tools is **non-destructive**: the original file on disk stays unchanged. RLMP stores the settings in the `.lmp` project file and applies them on the fly during playback.

To open a clip's settings, **right-click** on the card.

5.1 Basic properties

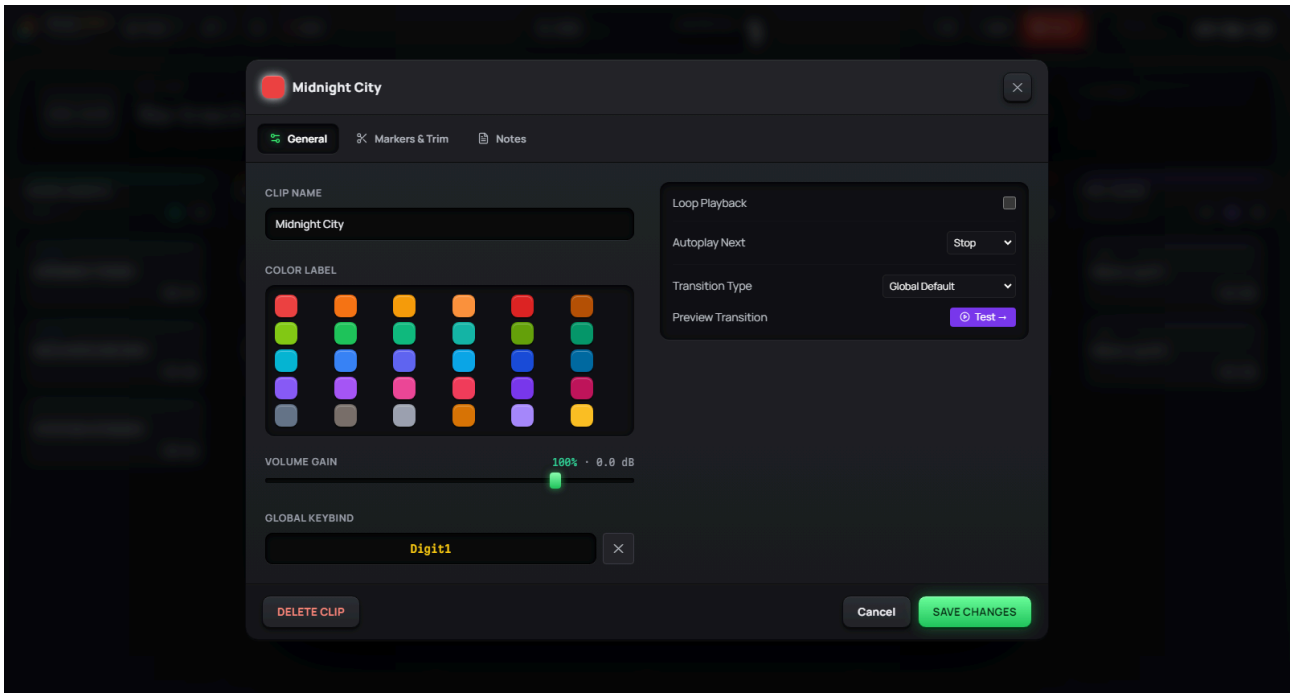


Figure 5.1 — The clip settings: name, colour label, Volume Gain, behaviour, Next Action and key assignment.

Name and appearance

Clip Name. You can give the clip a custom name, independent of the original file name. The name is displayed on the card in the grid. Use descriptive names that are operationally useful during the live show: "OPENING THEME" is easier to read than `theme_rev3_final_def.mp3` when you have three seconds to find the right clip.

Custom colour. By default, the clip inherits the colour of its column. Here you can assign a specific colour to make it stand out visually. Useful for marking critical clips (e.g. the closing theme) or for distinguishing thematic groups within the same column.

Volume (Gain)

The Volume Gain slider runs from 0% to 150% and acts as a pre-fader on the specific clip, before the global Master Volume.

The most common use case is level alignment: if you have a voice message recorded at low intensity (e.g. a WhatsApp message or a phone recording), you can push it beyond 100% to bring it closer to the volume of the other tracks. Conversely, you can lower a particularly “hot” clip without touching the Master Volume.

5.2 The waveform editor

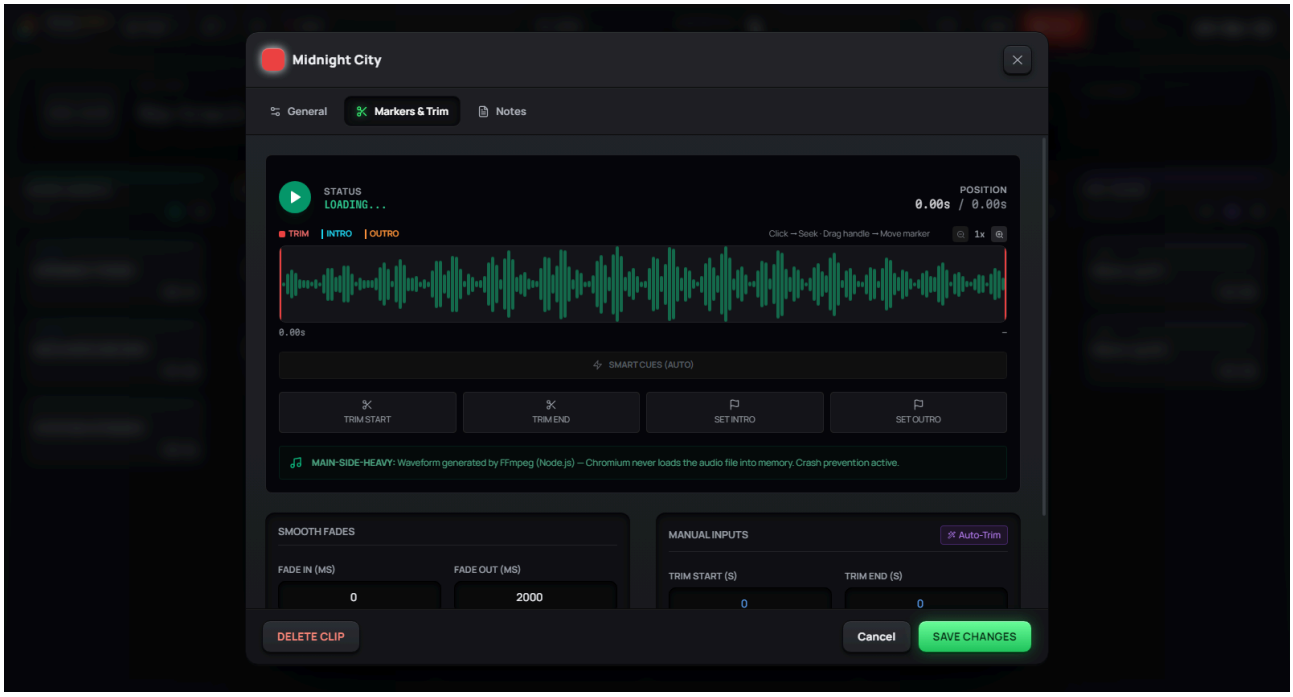


Figure 5.2 — The waveform editor: Trim handles, Intro and Outro markers, Auto-Trim, Smart Cues and fades.

The visual editor is the heart of the configuration panel. It occupies the central area and shows a graphical representation of the entire clip’s audio.

Navigating the editor

Horizontal zoom. You can zoom the waveform view from 1× (full view) up to 8×, with intermediate steps (1×, 2×, 3×, 4×, 6×, 8×), using the zoom slider or the mouse wheel over the editor. At high zoom, the view scrolls to follow the current position.

Adaptive ruler. The time axis at the top of the editor adjusts automatically to the zoom: at full view it shows sparse references, at maximum zoom it packs them down to the second.

Playhead. During preview playback, a white vertical indicator moves in real time along the waveform, showing the current position. A click on the waveform moves playback to that point.

The four handles

The editor has four draggable **handles**, each with a precise function and colour:

Trim Start (red handle, left). Defines the clip’s actual start point. Everything to its left is skipped during playback. Drag it right to remove silences or unwanted parts from the beginning.

Trim End (red handle, right). Defines the actual end point. Everything to its right is ignored. Drag it left to shorten the tail. Trim Start and Trim End cannot overlap.

Intro Marker (cyan handle). Marks the structural point where the main melody enters the track, after any intro. Once set, the countdown **INTRO −MM:SS** appears on the playing card.

Outro Marker (orange handle). Marks the point where the track's tail begins, typically the moment to start talking to fill the transition. The countdown **OUTRO IN −MM:SS** appears on the card. If the value turns out to be inconsistent with the trim or the duration, the software disables it and warns you.

Besides dragging, four Set buttons place each handle at the playhead's current position, for marking on the fly while listening. The values remain precisely editable in their respective fields.

Auto-Trim (Magic Wand)

The button with the **magic wand** icon starts automatic silence detection via FFmpeg. The threshold is not fixed: the software first estimates the file's average level and sets the silence threshold about 25 dB below it (within a safety range between −55 and −20 dB; if no estimate is available, it falls back to −40 dB). Trim Start and Trim End are set automatically, removing leading silences and mute tails without manual work.

This is especially useful for unprocessed voice recordings: phone calls, voice messages, interviews recorded on mobile devices. Running Auto-Trim over the entire Voice column before a show takes less than a minute and makes the transitions cleaner.

> **NOTE**
> The analysis happens in the Main Process via FFmpeg, without loading the file into the Renderer's memory. On large files, the analysis time stays in the order of a few seconds.

Smart Cues (automatic marker detection)

Alongside Auto-Trim, the **Smart Cues** feature automatically proposes the Intro and Outro markers. Using a more aggressive threshold, it finds the point where the audio reaches full energy (Intro) and the point where the final fade begins (Outro), placing the two markers without your having to find them by ear.

Transition preview

If there is a **next** clip in the same column, the **"Test →"** button plays the last few seconds of the current clip and lets the transition into the next one trigger, right there in the editor. During the preview a Stop button ends the test.

5.3 Behaviours and automation

Overlap: the column decides

When a clip is started, it interrupts any other clip playing in the same column (with a fade out): one song excludes the others. Only the **pad FX** effects overlap freely, playing on top of anything without stopping it.

If an element has to "ride" what's on air — a station ID ("You're listening to…") over a track's intro, a short jingle over a looping bed — no clip setting is needed: put it in the **pad FX** (it plays at full volume over the music) or in the **Voice** column (which additionally lowers everything else with ducking). A clip's purpose is determined by the column it lives in; the old per-clip "Stacco (Jingle)" behaviour of previous versions has been removed.

Next Action (end-of-clip automation)

Defines what happens when the clip reaches the Trim End point.

Stop — the default behaviour for Songs, Voice and Assets. The clip ends and stops.

Play Next — when the clip nears the end, it automatically starts the next clip in the column with the configured transition. The **NEXT** badge appears on the card. It's the default behaviour of the Pre-

Show column and effectively creates an automatic playlist: you can set it on several consecutive clips to build blocks that flow without breaks.

Loop playback is a separate option: when active, the clip restarts from the beginning (from Trim Start) seamlessly, and the **LOOP** badge appears on the card. Use it for music beds, sound environments or background idents that should keep running until explicitly stopped. The transition modes — Crossfade, Segue, Gapless — are described in Chapter 13.

5.4 Fades (Fade In and Fade Out)

The panel lets you set, per clip, the duration of the incoming and outgoing fades. Values range from 0 to 60,000 milliseconds (60 seconds) and the curve applied is linear.

Fade In. The time the volume takes to reach its maximum level from the start. A value of 2000 ms produces a gradual two-second rise. Use it on music beds that should emerge gently; keep it at 0 for voices and effects that need to be heard immediately.

Fade Out. The fade time at close, both when you click an active clip and during transitions. Typical values: 2000–3000 ms for songs, 500–1000 ms for beds, 0 ms for hard stingers.

A 0 ms fade out produces an immediate close (“hard cut”). On a music track, live, it can be perceived as a technical error: consider carefully when it is appropriate.

5.5 Assigning controls

Every clip can also be launched from a keyboard key or a MIDI controller.

Trigger Keybind. The keyboard key assigned to the clip. You can set it from the dedicated field in the clip settings (click and press the desired key) or from the **Keybinds** window reachable from the Tools menu. The corresponding badge appears on the card. If the key is already assigned to another clip, the software reports the conflict before overwriting.

MIDI Bind. The assigned MIDI note (e.g. `NOTE:60`). The assignment is made through **MIDI Learn** mode (see Chapter 8), not by typing the number by hand.

Clip bindings are saved in the project file: carry the project to another computer with the same MIDI controller and the mappings will work without reconfiguration.

CHAPTER 6

The mixing engine

The underlying problem of manual radio production is the multiplication of simultaneous actions: start a track, lower the music, speak into the microphone, prepare the next clip, keep an eye on the clock. Every extra operation is an opportunity for error, in a setting where the error is public and immediate.

Runtime Live Machine Pro's mixing engine eliminates most of these intermediate actions by delegating them to the software. It doesn't do things behind your back. It automates the rules you would apply yourself if you had enough hands to carry them all out.

6.1 The audio hierarchy

The automatic mixing system is based on a **priority hierarchy** among the types of clip. The easiest way to grasp it is to picture it as a scale of "the right to speak".

Voice / Recordings — absolute priority. When a voice clip is playing, it stays at its nominal volume and everything else drops. No other signal can override this rule.

Episode Songs. They yield space to Voice, but they rule over the Asset beds. When a song comes in, the Asset music beds go to zero (they don't stop: they keep running in silence, ready to return). This is Music Dominance, described further on.

Show Assets, Jingle and Promo — the service beds. They are lowered by Voice and silenced by Songs.

Pad FX effects. Sound effects stay outside the hierarchy: they play at their own volume, overlap whatever is on air, and are never silenced. There is one courtesy toward speech: when a voice is active, the effects drop to half volume (50%) so as not to cover it, then rise again on their own.

6.2 Automatic ducking

Ducking is the mechanism by which a signal is lowered when a higher-priority signal starts playing.

The most common case: a song is playing at full dynamics; you launch a pre-recorded interview from the Voice column. At that moment RLMP brings the song to about **20% of its volume** (a reduction of roughly 14 dB) with a soft half-second fade, so the voice occupies the sonic space intelligibly. As soon as the interview ends, the song rises back to its original volume with an equally smooth fade in.

The operator touches nothing. The gesture performed was a single click: starting the interview. The amount of the reduction and its speed are adjustable in the Settings (Chapter 13).

Smart Mic: the microphone that lowers the music

Ducking can also be triggered by your voice, without launching any clip. With **Smart Mic**, RLMP listens to the microphone and, when you speak, automatically lowers the music; when you stop, it brings it back up.

1. In the Settings, Microphone tab, enable **Smart Mic** and choose the input device (Chapter 13).

2. In the Control Bar, press **ARM**: the microphone icon is replaced by a small eight-LED VU showing the level of your voice.
3. Speak: when your voice goes above the threshold, ARM turns red and pulses, and the music drops; shortly after you stop, it comes back up.

> ⚠ **WARNING**
> Smart Mic is designed for a **USB microphone connected directly to the computer**. If the microphone goes through a USB mixer (Rødecaster, Zoom LiveTrak and the like), the computer's input also carries the music on air: detection mistakes it for your voice and triggers by itself. In that case leave Smart Mic off and let the mixer handle the ducking.

Smart Mic is off by default and the microphone stays disarmed until you press ARM: if you don't use the feature, nothing changes.

6.3 Music Dominance: intelligent management of beds

A classic sonic mistake is the moment when a song and a music bed overlap: two rhythmic elements colliding, two kick drums that don't line up, and the result is muddled.

RLMP handles this scenario with **Music Dominance**.

The typical scenario. A bed is looping in the Assets column, under the host's voice. The host launches a track from the Songs column.

What RLMP does. It doesn't stop the bed, because stopping it would then require restarting it by hand. Instead it silently brings it to **zero volume**, keeping it playing "as a ghost": the file keeps running, the loop continues, but nothing is heard.

The sonic result. Only the song is heard. The bed has vanished without the operator doing anything.

The return. When the song ends, the bed re-emerges with an automatic fade in, resuming from the point it had reached in the loop. The flow (bed → song → bed) happens without a single extra click.

6.4 Rising above the hierarchy: effects and voices

If an element has to rise above what's on air, there are two ways, and they depend on where you put the clip.

The **pad FX**: the effect plays at full volume over the music and stops nothing — ideal for stingers and short jingles that must "punch through" the mix without touching it.

The **Voice** column: the clip stays at its nominal volume and everything else is lowered with ducking — ideal for the spoken station ID ("You're listening to…") that has to be clearly audible while the bed underneath keeps running. For a more polished entrance, pair it with a short fade in (300–500 ms): the attack will be soft, not abrupt.

Previous versions had a per-clip behaviour called "Stacco" that reversed the hierarchy. It has been removed: a clip's position in the hierarchy depends solely on the column it lives in, with no exceptions hidden in its properties.

6.5 Volume levelling (loudness)

Clips from different sources almost always arrive at different levels: a properly mastered ident, a quietly recorded phone voice, a track downloaded at its own volume. To avoid constant manual

Gain adjustments, RLMP applies by default a **volume levelling** based on the EBU R128 loudness standard, with a target of **−16 LUFS**.

In practice, the software evaluates the perceived loudness of each clip and brings it closer to a common reference, so that songs, voices and beds start out on a coherent footing. The feature is enabled by default and the target value is adjustable in Settings → Master Chain.

6.6 Master Chain: the processor chain on the master bus

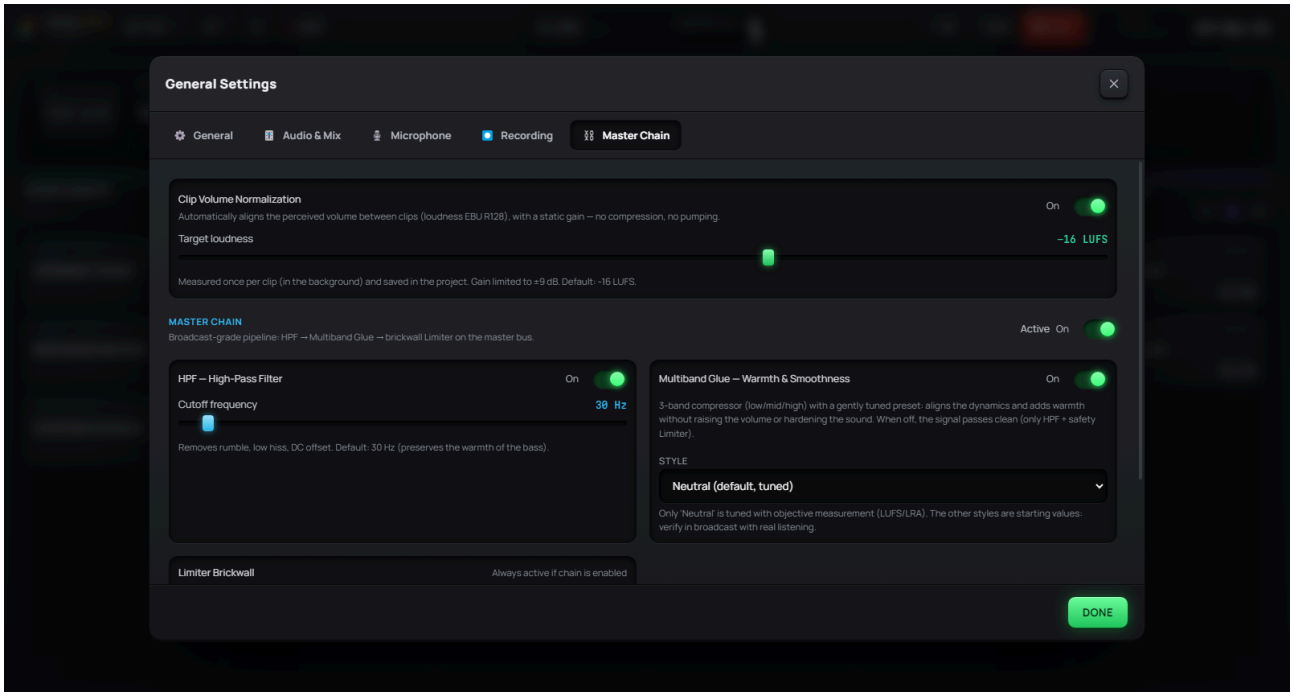


Figure 6.1 — The Master Chain: volume levelling (−16 LUFS), HPF at 30 Hz, multiband glue and brickwall limiter.

The combined signal of all playing clips, after the Master Volume, passes through a **processor chain** on the master bus before reaching the output device. The chain is enabled by default and designed for a broadcast-grade sound without requiring advanced configuration.

It comprises three stages in series.

High-Pass Filter (HPF) at 30 Hz. Removes the useless sub-bass frequencies that eat up headroom and can muddy playback systems, with a gentle slope. The cutoff frequency is adjustable (20–200 Hz). When disabled, the stage becomes completely transparent.

Multiband glue. Not a single compressor, but three “gentle” compressors working in parallel across three frequency bands (lows, mids, highs), separated by a crossover. Each band has thresholds and ratios calibrated to “glue” the mix without crushing it, and to hold the dynamic variance between clips of different levels in check. The style is selectable among a few presets (Neutral, Rock, Jazz, Electronic); the default preset is Neutral.

Brickwall limiter. Threshold at −1 dBFS, with a high limiting ratio and a very fast reaction. It guarantees the signal never exceeds the maximum allowed level, preventing digital distortion (clipping) whatever happens upstream.

The whole chain, and each individual stage, is configurable and can be disabled from Settings → Master Chain, where you’ll also find a button to restore the defaults. In a context where the signal is already processed by a hardware mixer or an external chain, you can disable it to avoid double processing.

CHAPTER 7

The Pad FX and the Automix view

Two working surfaces live above the grid, each summoned with a key and built for opposite moments of production. The **pad FX** fires off effects and stingers reliably without interrupting anything. The **Automix view** handles the music flow the way a DJ would. Neither takes space away from the grid: they open when you need them and close with a click.

7.1 The pad FX: the jingle machine



Figure 7.1 — The pad FX: the 5×5 jingle machine of sound effects, with overlapping playback.

Sound effects have no column in the grid. They live in the **pad FX**, a grid panel of cells (a jingle machine) that opens from the **FX** button in the header (the button's LED turns green) and stays floating in a corner of the screen.

The pad is a **non-blocking overlay**: it doesn't obscure the board and doesn't intercept clicks elsewhere. You can launch an effect and, at the same moment, keep operating on the columns or the header controls. For that reason the `Esc` key doesn't close the pad: it remains the STOP ALL command, always available. The pad closes from its own close button or by toggling FX again.

Loading and launching effects

The pad starts with a grid of 25 cells (5×5) and grows in rows as you add more effects. To populate it, **drag audio files straight onto the pad's cells**, exactly as you would with a grid column.

A click on a cell **launches the effect**. The pad effects are polyphonic and overlap: several cells can play together, over anything on air, without stopping it. The audio behaviour is identical to a normal clip; only the launch surface changes. The cell of the effect on air lights up in its colour, and a counter on the FX button in the header shows how many effects are playing at that moment.

Configuring an effect

Effects are configured on two levels, meant for two different needs:

- **Quick settings** — the common case for a jingle machine: name, colour, volume, loop. It takes a few seconds.
- **Full settings** — the same window as the grid clips (waveform editor, trim, markers, fades, key assignment), reachable from the “Full settings…” entry within the quick settings.

Pad position

The pad can sit in the bottom-left or bottom-right corner of the screen: the preference is set with the arrow button in the pad’s header and is remembered across sessions. On the right it covers the NoteBoard and the last column; choose the side based on how you’ve laid out your running order.

> **NOTE**
> In MIDI Learn mode, a click on a pad cell **selects** the effect for assignment instead of playing it — so you don’t send a jingle on air while mapping the controls (see Chapter 8).

7.2 The Automix view

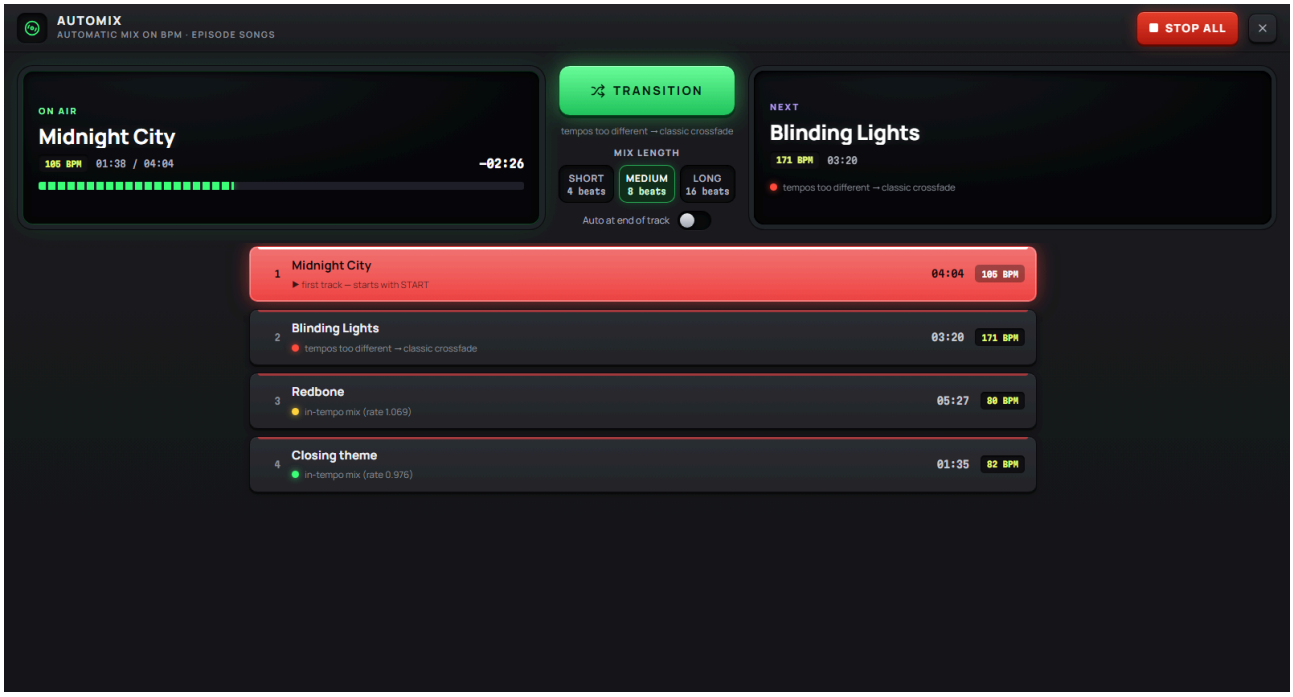


Figure 7.2 — The Automix view: the Music-column deck, the mix length (short, medium, long), BPM compatibility and automatic end-of-track mode.

The **Automix view** is the deck of the Music column: a full-screen display, summoned from the **MIX** button in the header, that presents the music running order like a DJ console. It opens above the board but below the pad FX, so the effects stay usable even while Automix is open. As with the pad, `Esc` doesn’t close it: it remains the emergency command, and the STOP ALL button stays reachable in the header.

Opening the view turns on **MIX mode**: while it stays open, **every change from one Music-column track to another is a mix**, even when you make the change yourself. Closing the view brings back the normal behaviour, where a track launched by hand cuts the one on air.

The deck

At the top there are two displays. On the left, the track **on air**: title, BPM, elapsed time and duration, time remaining and an LED progress bar. On the right, the **next** track in the Music column, with BPM,

duration and compatibility. Between them sits the big green **TRANSITION** button, which moves to the next track with a mix. If nothing is on air, you find **START** in its place, which starts the first track.

Below the displays is the Music column's running order, one row per track with number, title, duration and BPM. The row of the track on air is lit in the column's colour.

Moving from one track to another

With the view open you can change track in four ways, and all of them produce a mix:

- **TRANSITION** — moves to the next track in the running order. Tracks whose file cannot be found are skipped.
- **Clicking a row** of the running order — moves to that track, wherever it sits. It works from the keyboard too: with the row selected, `Enter` or `Space`. If nothing is on air, the click starts the track. The rows of the track on air and of the one going out don't respond, so a careless click cannot stop them.
- **Keys and MIDI notes** assigned to Music-column tracks (Chapter 8).
- **From the phone**, with Remote Control: the **NEXT** button and the buttons of the single tracks (Chapter 11).

The track you are leaving fades out while the new one comes in. If you move on before the fade has finished, the oldest track leaves within a quarter of a second: there are never three tracks on air on top of each other.

Mix length

Below the transition button you choose how long the change lasts:

- **SHORT (4 beats)** — about two seconds at 120 BPM. It's the choice for fast changes: a **chart countdown** where only the start of each track plays, or a **medley** while waiting for the televote to close.
- **MEDIUM (8 beats)** — about four seconds at 120 BPM. This is the default.
- **LONG (16 beats)** — for unhurried music blocks.

The length is counted in beats of the track on air, so the change always lasts the same number of beats. Your choice is remembered. When the change is a classic crossfade, its length is the crossfade length set in the Settings, Audio & Mix tab (Chapter 13).

Below the buttons, a line reports the **outcome of the last change**: beat-matched mix in green, or classic crossfade with the reason, or not carried out, in amber.

Compatibility and beat-matched mixing

Below the title of each track, a **compatibility dot** with the previous track shows their rhythmic affinity, followed by a short explanation (for example "beat uncertain → classic crossfade"):

- **Green** — the two tempos lock well: the change will be a beat-matched mix.
- **Yellow** — locking is possible, with a more noticeable speed adjustment.
- **Red** — the change will be a classic crossfade; the explanation says why.

In a **beat-matched mix** the incoming track starts on one of its beats, lined up with the beats of the track on air, and its speed is adjusted (by at most 8%, with the pitch preserved) so that the two tempos match. Once the mix is over, the speed returns to the original one little by little, too slowly to be heard. Even in fast changes the new track locks onto the tempo you actually hear, including that of a track that has just come in and is still sped up or slowed down.

Tempo and beat positions come from an automatic analysis of every track in the Music column, done when you drag it into the column or when you open the project. The analysis also gives a **beat-reliability score**. Songs with a steady kick drum pass it easily; ballads, tracks without drums or with a changing tempo usually don't.

When rhythmic locking isn't practical (BPM not detected, unreliable beat, tempos too different), the software declares it and falls back to a **classic crossfade**. The worst outcome is always a normal fade, never beats locked at random.

> **NOTE**
> From version 1.15.37 beat detection is more accurate. The first time you open a project created with an earlier version, the tracks in the Music column are analysed again, once only and in the background. Until a track's analysis has finished, changes involving it are classic crossfades.

> **TIP**
> Chart countdown or medley: open the MIX view, choose **SHORT**, press **START** and use **TRANSITION**, or click the track you want directly. From the phone you do the same with **NEXT** (Chapter 11).

Automatic mode

Below the transition button there's the **Auto at end of track** switch. When it's active the label turns green and RLMP starts the passage to the next track on its own when the track on air nears the end.

This mode is a deliberate exception to the software's philosophy, which by design does not automate the show. That's why it is **disabled by default** and works **only while the Automix view is open**: closing the view disables the automation. It's the right tool for a continuous music block, the half-hour of music-only before returning to the mic, not for the entire live show.

CHAPTER 8

Hardware, keyboard and MIDI

Runtime Live Machine Pro is designed to integrate with the hardware already in the studio without requiring elaborate configurations. This chapter describes how to route the audio output, how to use the computer keyboard as a controller, and how to connect physical MIDI devices for tactile control of the production.

8.1 Audio routing

Selecting the output device

By default, RLMP outputs to the operating system's default audio device. In a professional or semi-professional setting, with USB mixers, external sound cards or multitrack systems, it's useful to select the signal destination explicitly.

1. Open the **Settings** from the Tools menu.
2. On the Audio & Mix tab, open the output-device menu: you'll find the list of audio devices available on the system.
3. Select the device you want.

If the chosen device is unplugged, RLMP falls back automatically to the system default; the app monitors connections and reacts to the insertion or removal of USB devices.

USB mixers and multichannel setups

USB mixers such as the Rødecaster Pro, the RØDECaster Duo or the Focusrite Scarlett typically expose several USB channels to the operating system (Main Mix, Sounds/Chat, Monitor, and so on). RLMP appears as a single stereo source; the choice of which USB channel to route it to is in your hands.

Recommended setup with a USB mixer. Assign RLMP to a secondary channel of the mixer (e.g. "Sounds" on the Rødecaster Pro) rather than to the main channel. This way you control RLMP's volume with a dedicated physical fader, keep it separate from the physical microphone signal, and apply any hardware processing to that channel only.

Latency and buffer

RLMP uses the operating system's native audio APIs. The output latency is determined by the audio device's buffer, not by the software. With professional sound cards the latency is in the order of a few milliseconds, imperceptible in a playout context.

If you notice audio artefacts (crackles, dropouts), the device's buffer value is probably too low. Raise it from the sound card's control panel (not from RLMP, which doesn't manage the driver directly): a buffer of 256 or 512 samples is the ideal balance between latency and stability.

8.2 Keyboard control

The computer keyboard is the fastest controller available on air: it works in the dark and is always within reach. RLMP provides a set of global shortcuts and lets you assign keys to individual clips.

Global shortcuts

Key	Action
Esc	STOP ALL — stops all active clips
Delete / Backspace	Delete the selected clips
Ctrl+Z	Undo the last change to the running order
Ctrl+Y (or **Ctrl+Shift+Z**)	Redo the undone change
Ctrl+Shift+D	Show/hide the Debug Overlay
Ctrl+Shift+M	Open the MIDI simulator (for testing without a controller)
F1, F2, F3…	Launch the matching column, counting the visible columns from the left

`Esc` acts as STOP ALL when RLMP is the active window, even while the cursor is in a text field. It is no longer a shortcut registered at the operating-system level: if the app is in the background, bring the window to the foreground first.

The other shortcuts are suspended while a window is open (Settings, the clip editor and the like), so `Ctrl+Z` or `Delete` don't touch the running order while you work elsewhere. With a window open, `Esc` closes it without stopping the live show.

Function keys: one column per key

The **F1**, **F2**, **F3**… keys launch the first available clip of the columns **visible** in the grid, in the order you see them from left to right. With the default layout: F1 Show Assets · F2 Jingle · F3 Promo · F4 Episode Songs · F5 Voice · F6 Pre-Show. If you hide a column from the Settings (Chapter 13), the keys shift accordingly: F1 is always the first column you see. If you have assigned a function key to a specific clip, that assignment takes precedence.

> **⚠ WARNING**
> F1 normally launches Show Assets, and a clip from that column launched by hand stops everything on air except the pad FX effects, exactly like clicking the same clip.

Custom keys per clip

In addition to the global shortcuts, every clip can have a dedicated key. The corresponding badge appears on the card.

To assign a key: 1. Open the clip settings (right-click the card) or the **Keybinds** window from the Tools menu. 2. Click in the key field. 3. Press the key you want.

Available keys. Almost any key: letters (A–Z), numbers (0–9), numeric keypad, spacebar, unused function keys. If the key is already assigned to another clip, the software reports the conflict before overwriting, so you don't create invisible duplicates.

Safety while typing. Custom keys are disabled automatically when you're in text-entry mode (renaming a clip or writing a note). This prevents accidental launches while you type.

8.3 MIDI controllers

MIDI is the professional choice when you want physical, reliable control under your fingers. RLMP supports USB-MIDI controllers: keyboards, pads (e.g. Novation Launchpad), fader controllers (e.g. Korg nanoKONTROL2), hybrid control surfaces.

Connecting

Connect the USB controller to the computer and start RLMP. The software detects devices through the system's Web MIDI API and recognizes the connection and disconnection of a controller in real time. Most USB-MIDI controllers are class-compliant and need no driver; for professional surfaces with proprietary drivers, install the driver before connecting the device.

MIDI Learn

RLMP doesn't require you to know MIDI note numbering or to configure messages by hand. Learning is done through **MIDI Learn** mode, from the Tools menu (or from the Keybinds window).

To map a clip to a key/pad: 1. Enable MIDI Learn. The cards enter a waiting state, with a dashed cyan border. 2. Select the clip (or the pad FX cell) to map. 3. Play the note, press the pad or the key on the controller. The `M` badge with the note number appears on the card.

To map the global functions: - Select **STOP ALL** and press a key on the controller: that key will perform the Stop All. - Select the **Master Volume** and move a fader or a knob: that control will manage the master volume continuously.

When finished, disable MIDI Learn to return to operating mode.

Supported message types

Note On — messages generated by buttons, pads and keys. Ideal for launching clips and global actions; RLMP responds to the key press and recognizes all MIDI channels. Note Off messages are ignored.

Control Change (CC) — messages generated by faders and potentiometers, with a continuous value from 0 to 127. Ideal for the Master Volume: a physical fader mapped to the master offers the most natural control of the output level.

Portability of the mappings

The MIDI mappings of the **clips** are saved in the `.lmp` project file: carry the project to another computer with the same controller and they will work without reconfiguration. The mappings of the **global functions** (Stop All, Master Volume) are instead tied to the computer, saved in the application's local preferences, and remain valid for all projects on that machine.

CHAPTER 9

Session recording

Session recording turns Runtime Live Machine Pro from a playout tool into a complete production tool. You don't need separate recording software or a virtual routing chain. RLMP captures the **post-processed master mix** directly, everything that leaves the application (including the Master Chain effects), into an audio file on disk.

9.1 Starting the recording

The recording control is the **REC** button, with a red circle, in the left zone of the header, next to the LOG.

Start. Click **REC**. The button turns red and shows the recording time running, next to a flashing stop icon: capture is under way. Recording starts immediately: everything that leaves the software's output from that moment on is captured.

You don't need clips playing to start recording: you can begin capture ahead of the show's start, so you don't lose the first few seconds in case of an early start.

What gets recorded. The captured signal is the **master after the limiter**: it includes the mix of all playing clips and the processing of the entire Master Chain (HPF, multiband glue, limiter). It is exactly the signal that reaches the audio output device.

The internal format. During capture, RLMP writes a compressed Opus stream (in a WebM container) at 320 kbps: very light on disk and transparent to the ear. Continuous recording has a safety limit of about **four hours**; beyond that duration the capture stops automatically so as not to saturate memory.

System overhead. Capture happens downstream of the audio engine, without burdening the Renderer. You can record sessions hours long without worrying about resource consumption.

9.2 Stopping the recording and choosing the format

When you click the button again to stop the recording, the **export window** opens. This is where you choose which format to save the file in: converting from the internal stream to the final format is handled by FFmpeg.

Available formats

Format	Extension	Characteristics
WAV	`.wav`	Uncompressed lossless. Maximum quality, large files. Ideal for archiving and post-production.
FLAC	`.flac`	Compressed lossless. Same quality as WAV, smaller size. Ideal for archiving.
MP3	`.mp3`	Lossy. Selectable bitrate. Ideal for distribution and podcasts.
OGG	`.ogg`	Open-source lossy. Good quality-to-size ratio.
WEBM	`.webm`	Lossy, optimized for the web. Matches the internal capture format.

Quality options

For the lossless formats (WAV and FLAC) you can select the **bit depth**: 16-bit (CD standard), 24-bit (professional broadcast standard, the default) or 32-bit float (maximum precision, if the recording will be mastered later).

For the lossy formats (MP3, OGG, WEBM) you can select the **bitrate** among 128, 192, 256 and 320 kbps. For a podcast intended for online distribution, 192 kbps stereo is the recommended minimum; 256 kbps is the current standard for "transparent" quality.

Choosing the save location

In the export window you choose the destination folder and the file name. If you don't specify a name, RLMP generates one based on the session's date and time. When the conversion finishes, a confirmation toast shows the path of the saved file.

9.3 Practical considerations

Synchronizing with the show

The recording captures all the time elapsed between Start and Stop, including the silences. If you started capture 30 seconds before the show's actual start, the resulting file will include those first 30 seconds. For a distribution-ready result without post-editing, start the recording exactly when the show begins.

Recording and backup at the same time

The project's autosave system (see Chapter 10) and session recording operate independently. You can record a show while the autosave silently saves the project state: the two operations don't interfere.

Recommended format for different contexts

Podcast — MP3 256 kbps stereo or FLAC 16-bit. The first if you distribute the file directly, the second if it will pass through an editor.

Long-term archive — WAV 24-bit or FLAC 24-bit. Generous sizes, maximum flexibility for any future remasters.

Radio / Streaming — check your platform's requirements. Most accept MP3 128–192 kbps; some require uncompressed WAV. RLMP exports in the most common formats to cover every scenario.

CHAPTER 10

Project management and data safety

Preparing a show takes time: selecting the files, organizing them in the columns, configuring the volumes, setting the fades, assigning the keys. That work is an operational asset that must survive any mishap: a system crash, a move to another computer, the return to an episode archived months earlier.

RLMP addresses data safety on several levels, each designed to cover a specific risk.

10.1 The project file (.lmp)

The entire state of a show (the layout of clips in the columns, the custom names, the volumes and fades, the editor cue points, the NoteBoard notes, the MIDI and keyboard mappings, the column colours) is saved in a file with the `.lmp` extension (Live Machine Project).

The format is JSON: a structured text file, readable by any editor, non-proprietary. If RLMP were one day unavailable, the project data would remain accessible.

What the `.lmp` file contains: all the settings listed above, including the absolute paths to the referenced audio files.

What it doesn't contain: the audio files themselves. The `.lmp` stores where the files are on disk, it doesn't copy their content. A project file is typically in the order of kilobytes, regardless of how many or how large the audio files it references are.

On opening, RLMP validates the file: it rebuilds any duplicate identifiers, brings out-of-range values back within sane limits, and, if you open a project created with an earlier version, automatically adds the columns introduced in the meantime (Jingle, Promo) without touching the existing data.

10.2 Saving

Quick save

The Save Project entry in the FILE menu performs an immediate save to the open `.lmp` file. The save is silent: no dialog box. When there are unsaved changes the entry turns amber and an amber dot lights up on the FILE button, a reminder you can see at a glance. Use it often while preparing the show.

The save is **atomic**: the file is first written to a temporary copy and then renamed on the fly. If the computer shuts down during the write, the original `.lmp` is never left half-written.

Save As

The Save As... entry always opens the dialog box, even if the project already has a name. Use it to:

- Create progressive versions of the same show (`Ep47_draft.lmp` , `Ep47_v2.lmp` , `Ep47_final.lmp`).
- Save a variant with different configurations.
- Create a new file without overwriting the current one.

Protection on close

RLMP continuously monitors the state of changes. If you try to close the software (or open a new project) with unsaved changes, the operation is suspended and a confirmation request appears with three choices: save, discard the changes, or cancel. It is not possible to lose work through an accidental click on the window close.

10.3 Auto-Backup and autosave

Beyond the saves you decide on, the software maintains an automatic safety net.

Project backup copy. Every time an already-saved project is updated in the background, RLMP keeps a `.bak` copy alongside the `.lmp` with the last valid state.

Rotating autosave. In parallel, RLMP writes snapshots of the current state to a dedicated application folder, `autosaves`, with a name based on date and time. The **ten most recent snapshots** are kept: the oldest are deleted as new ones are made. This net also captures work on an "untitled" project that was never saved to disk.

The `autosaves` folder is in the application data directory:

- **Windows:** `%APPDATA%\runtime-live-machine-pro\autosaves\`
- **macOS:** `~/Library/Application Support/runtime-live-machine-pro/autosaves/`
- **Linux:** `~/.config/runtime-live-machine-pro/autosaves/`

How to recover. If the main `.lmp` file has become corrupted or the computer shut down unexpectedly, open the `autosaves` folder, find the snapshot with the date and time closest to the moment of the interruption, and load it in RLMP like a normal project file. Alternatively, rename the `.bak` file next to the project to `.lmp` and open it.

10.4 Export project with audio

Because the `.lmp` file contains only the paths to the audio files, not the files themselves, a project is fragile: if you move, rename or delete even a single one of the source files, the corresponding clip turns red. The **Export project with audio** feature, in the FILE menu (right below Save As...), solves the problem at the root by consolidating all the audio inside the project.

How it works

RLMP analyses all the audio-file paths in the project, creates an `audio/` subfolder next to the `.lmp` file, and **physically copies** every referenced file into it. Files already present and identical are not re-copied; any name duplicates are renamed so they don't overwrite each other, and orphan files (no longer referenced) are removed from the folder.

The difference from a simple backup is what happens **after** the copy: RLMP **repoints every clip to the new copy** inside `audio/` and **re-saves the project**. From that moment on, the `audio/` folder is not a spare archive sitting next to the project, but the source the session actually reads its audio from.

The result: you can delete the originals

Because the project now points to the copies in `audio/`, **the audio files in their original location are no longer needed** and you can safely delete them: the show keeps working by reading from the archive. This is the difference from earlier versions, where the `audio/` folder stayed an orphan duplicate and deleting the originals broke the clips.

The project folder thus becomes self-contained: the `.lmp` plus the `audio/` subfolder, everything needed to run the show, ready to archive, copy, or carry to another computer with RLMP installed.

A few useful details:

- The operation is **repeatable**: if you add new clips and export again, RLMP copies only the new files and realigns the project, without duplicating the ones already archived.
- RLMP **only deletes files inside `audio/` folders it created itself**: the first export leaves a small text file, `.rrlmp-archive`, in the folder to mark it as an archive. If an `audio/` folder of your own already sits next to the project (or one created by versions up to 1.15.16), RLMP removes nothing: it tells you how many files are not part of the project and marks the folder, which it will manage normally from the next export. Move elsewhere anything you want to keep. If you delete `.rrlmp-archive`, RLMP stops treating the folder as its own.
- Repointing to the archive **does not enter the Undo/Redo history**: an Undo would send the clips back to the originals, which you may already have deleted.
- The project folder (`.lmp` + `audio/`) can be **moved, renamed, zipped and taken to another computer**: on opening, any file that isn't found at its old location is automatically searched for in the `audio/` folder next to the `.lmp` and relinked by itself. The project is marked as modified: the next save consolidates the new paths.

> **TIP**
> Use Export project with audio at the end of preparing each show to consolidate the audio inside the project. You'll have a compact, portable "master", and you can free up space by deleting the scattered files you imported from.

Integrity check on opening

Every time you open a `.lmp` file, RLMP runs an automatic **integrity check**: it verifies that each referenced audio file is reachable. Clips whose file is missing are flagged by a dark red card with the MISSING FILE badge. The rest of the project — all the clips with reachable files — stays fully functional.

CHAPTER 11

Remote Control

The person presenting isn't always seated at the computer. Sometimes the host is on the other side of the studio, behind glass, or moving around with a guest. The **Remote Control** in Runtime Live Machine Pro turns a phone, a tablet or a second computer into a small console linked to the studio: the **effect pads** (applause, laughter, gong…), the **Music column** with **MIX mode**, and the two stop buttons, **STOP FX** and **STOP ALL**. All it takes is the browser: nothing is installed on the remote device.

11.1 How it works

When you enable it, RLMP starts a small **local web server** inside itself. The remote device opens an address in its browser and gets a control page that mirrors the studio in real time: whatever starts or stops on the main computer shows up on the phone at once, and vice versa.

Everything happens **inside the local network**: the server is reachable from devices on the studio's own Wi-Fi or wired network, and it doesn't go through the internet. The server listens on port **8787**.

11.2 Activation

1. Open the **Settings** from the Tools menu and stay on the General tab.
2. In the **Remote Control** section, set the switch to **Active**.
3. A panel appears with the **QR code**, the six-digit **PIN**, the **address** to open on the remote device and the number of **connected devices**.

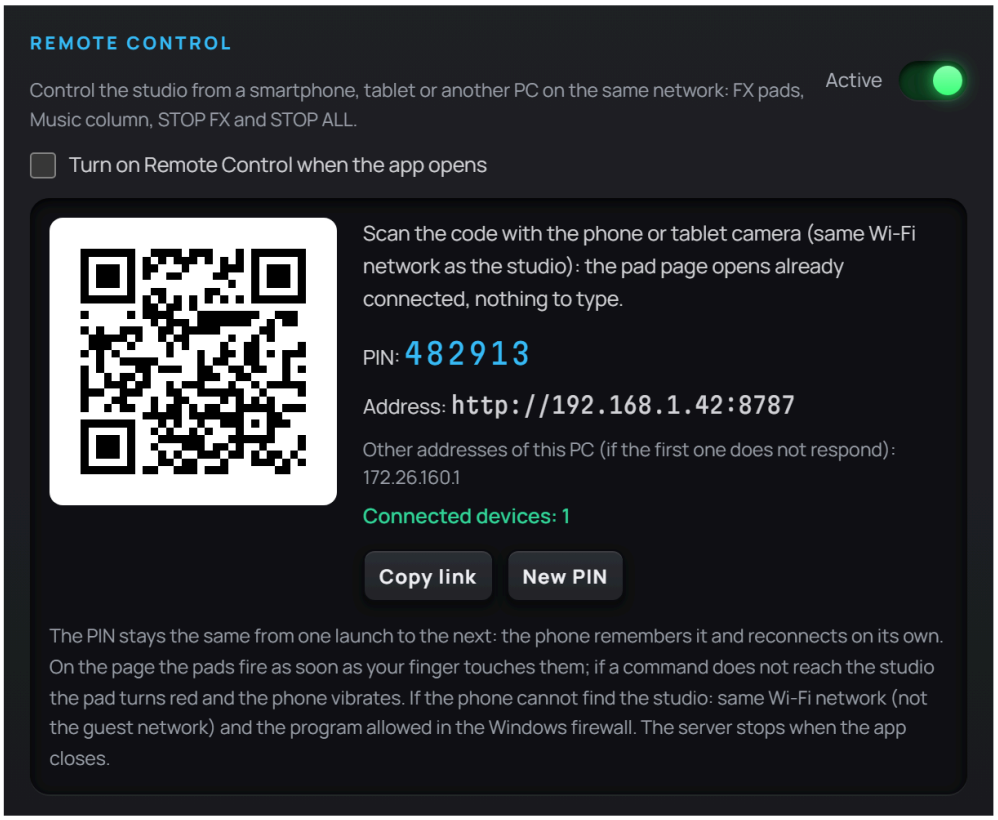


Figure 11.1 — The Remote Control section of the Settings: QR code, PIN, address, connected devices, “Copy link” and “New PIN”.

The panel contains:

- **QR code.** Scanned with the phone’s camera, it opens the control page already connected: there is no address and no PIN to type. The PIN travels in the part of the link after the `#` sign, which the browser never sends over the network.
- **PIN.** Six digits. **It stays the same from one launch to the next**: a phone that has used it once reconnects on its own, even after RLMP has been closed and reopened.
- **Address.** In the form `http://<computer-address>:8787`. RLMP lists the address of the real network (Wi-Fi or cable) first and those of virtual adapters (WSL, Hyper-V, VirtualBox) last. If the computer has more than one, the **other addresses** appear below, to try if the first one does not respond.
- **Connected devices.** Updated live, and green when at least one device is connected: it confirms the phone is linked to the studio without having to try a pad on air.
- **Copy link.** Copies the address (without the PIN) to the clipboard, for example to send it to your phone via Telegram or WhatsApp.
- **New PIN.** Generates a different PIN, for example after showing it to someone. Connected devices are disconnected and must enter the new PIN or scan the QR code again; if at least one is connected, RLMP asks for confirmation.

Below the switch there is the **Turn on Remote Control when the app opens** box. It is **off by default**: in that case the server stays off until you turn it on by hand, and no network service runs without your say-so. Tick it if you use the phone in every show.

If startup fails, typically because port 8787 is already taken by another copy of RLMP left open or by another program, the switch turns back off and a message shows the reason.

11.3 Connecting the phone or tablet

The quickest way is to **scan the QR code** with the phone's camera and tap the link that appears. The page opens already connected.

Otherwise, open the device's browser, type the address shown in the Settings and enter the **PIN**: the page connects by itself as soon as the sixth digit is in. From then on the PIN is remembered on the device, and next time you only need to reopen the page.

The page follows the **language of the device's browser**: Italian or English.

At the top there is the status line, with a coloured dot and a label: **Connected** in green, followed by the connection delay in milliseconds (round trip). On the right there are two buttons:

- ⛶ **(full screen)** — hides the browser bar and leaves all the space to the controls. It doesn't appear where the browser doesn't allow it, for example on iPhone.
- **PIN** — forgets the stored PIN and goes back to the PIN screen. Use it to connect to another studio or with another PIN.

> **TIP**
> On a local network the browser cannot stop the phone from turning its screen off. The page reconnects by itself when the phone wakes up, but on air it's worth lengthening the screen timeout, or keeping the phone on charge.

11.4 The FX tab: the pads from your phone

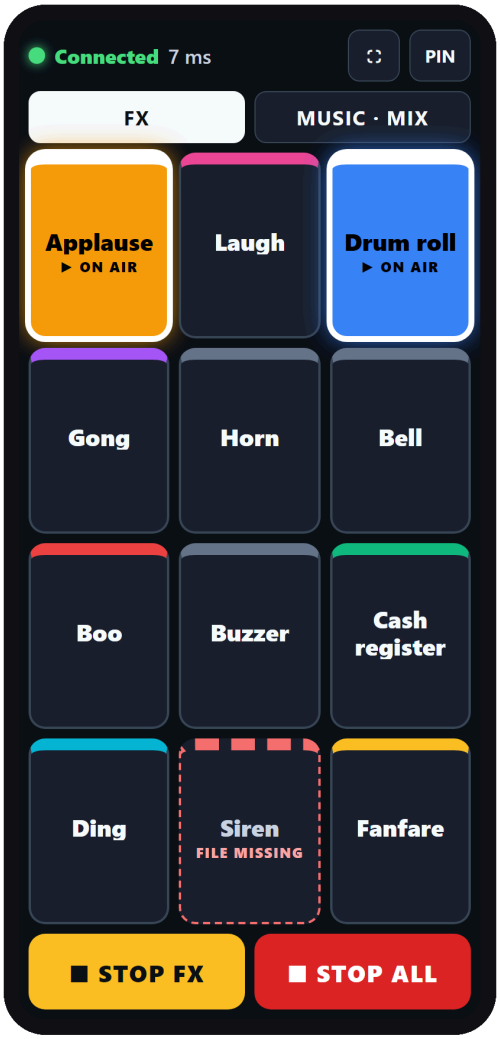


Figure 11.2 — The FX tab on the phone: two effects on air, lit in their colours, and STOP FX and STOP ALL at the bottom.

The **FX** tab is the one that opens first. It shows every effect in the studio's pad FX (Chapter 7), with the same names and colours, laid out in a grid that fills the screen with no scrolling. If there are many effects, they are spread over pages that you turn with the ◀ and ▶ buttons.

- **A pad fires as soon as your finger touches it**, not when you lift it, like a hardware pad: the time between the tap and the effect on air is the network's, usually a few thousandths of a second.
- **Tapping again** an effect that is on air stops it, exactly like clicking the pad in the studio.
- **Several effects play at once**: applause does not stop laughter.
- An effect on air **lights up in its colour** with the label **▶ ON AIR**, and goes dark by itself when it ends.
- An effect whose file cannot be found has a red dashed border and the label **FILE MISSING**: tapping it does nothing.

Every tap is confirmed by the studio. If the confirmation doesn't arrive within a second and a half, the pad turns red with the label **NOT RECEIVED** and the phone vibrates (where vibration is available): you know at once that the effect did not start.

At the bottom of the page there are two large buttons:

- **■ STOP FX** — stops only the effects. The music and everything else stay on air.
- **■ STOP ALL** — stops everything, like the ALL button or the `Esc` key in the studio.

Unlike the pads, these two buttons fire when you **lift** your finger: they sit where your thumb rests, and a STOP ALL fired by mistake would leave the show silent.

11.5 The MUSIC tab and MIX mode



Figure 11.3 — The MUSIC tab with MIX on: NEXT with the title of the following track, the track GOING OUT during the mix and the one ON AIR.

The **MUSIC** tab mirrors the studio's Music column. At the top are the **MIX mode** controls (Chapter 7):

- **MIX ON / MIX OFF** — turns MIX mode on and off. Turning it on from the phone also opens the MIX view on the studio screen. Next to it, a line reminds you what changing track does: "beat-matched mix" with MIX on, "hard cut" with MIX off. While MIX is on, the tab is called **MUSIC · MIX**.
- **▶▶ NEXT** — the large button. It shows the title of the next track and how the change will happen: **beat-matched mix** or **classic crossfade**. Tapping it, the studio moves to that track with a mix, even if MIX is off: it's the command for fast changes. Tracks whose file cannot be found are skipped. If nothing is on air, the button becomes **▶ START** and starts the first track; once the playlist is over it becomes **END OF PLAYLIST** and does nothing.
- **Last change** — a line tells you how the last change went: beat-matched mix (in green), classic crossfade or not carried out (in amber).

Below is the track list. The track on air carries the label **▶ ON AIR**; during a mix, the one going out carries the label **GOING OUT** and an amber border. Each row has a button:

- ▶ **PLAY** — starts the track. With MIX on and a track on air it becomes **MIX ▶**: the studio gets there with a mix instead of a cut.
- ■ **STOP** — stops the track that is on air or going out.

STOP ALL stays at the bottom of the page; STOP FX only appears in the FX tab.

> **TIP**
> For a fast chart countdown, or for a medley while waiting for the televote to close: in the studio choose the **SHORT** mix length in the MIX view (Chapter 7), then on the phone turn MIX on, press ▶ **START** and, whenever you want to move on, ▶▶ **NEXT**. Your hands stay free for the microphone.

11.6 Connection and status

The page keeps an eye on the connection by itself. Every three seconds it checks that the studio answers and shows the delay in the status line. If something goes wrong, it says so:

- **Reconnecting…** (amber) — the connection dropped and the page is trying again. The pads turn grey until it's back.
- **Not connected — retrying…** — the studio doesn't answer (computer off, no network, RLMP closed). The page keeps trying and reconnects as soon as it can, without asking for the PIN again.
- **Remote control turned off in the studio — waiting…** — someone turned Remote Control off in the studio. The page reconnects when it is turned back on.
- **Not connected: the command was not sent.** — you tapped a pad while there was no connection.

Taps made without a connection **are not queued**: applause that starts ten seconds late is worse than applause that never starts.

When you turn the phone's screen back on or return to the page, the connection is checked at once. If a **new PIN** has been generated in the studio, the page goes back to the PIN screen with the message "The PIN was changed in the studio: enter the new one". If the page is already open, just scan the new QR code: it reconnects by itself.

11.7 Security and limits

- **PIN required.** No device can send commands without the six-digit PIN. The PIN is created with the system's cryptographic generator.
- **Attempt protection.** After ten wrong attempts in five minutes from the same device, access from that device is temporarily blocked.
- **Only from the remote control page.** The server accepts connections only from the page it serves itself: a web page from another site, opened on a device on the same network, is rejected before it can even ask for the PIN.
- **Closed list of commands.** The server accepts only the intended commands: firing and stopping FX pads, STOP FX, starting and stopping tracks in the Music column, NEXT, turning MIX mode on and off, STOP ALL. Every pad or track commanded must exist in the studio; any other request is refused.
- **What stays in the studio.** The other columns, the clip editor, the settings, recording and project management are used only from the main computer.
- **Local network only.** The server is meant for the studio network. The PIN doesn't change by itself: if the network is open or shared, consider who can reach it and use **New PIN** when needed.
- **Shutdown.** The server stops when the application closes.

11.8 If the phone cannot find the studio

1. **Same network.** Phone and computer must be on the same Wi-Fi network, **not the guest network**: many routers' guest networks isolate devices from each other.
2. **Windows firewall.** The first time the server starts, Windows asks whether to allow Runtime Live Machine Pro on the network. Allow it. If your home or studio network is set as "Public", allow it on public networks too.
3. **Another address.** If the first address doesn't respond, try the others listed in the Settings.
4. **Port in use.** If the switch turns itself off with a message, port 8787 is used by another copy of RLMP or by another program: close it and turn Remote Control back on.

CHAPTER 12

Updates

Runtime Live Machine Pro updates itself, but never at your expense. Two rules govern everything: no update may interfere with a live show, and no download starts without your consent. This chapter explains how the software checks for new versions, how it installs them, and why it sometimes behaves differently depending on the operating system.

12.1 The startup check

Shortly after startup (about three seconds), RLMP silently checks whether a newer version exists. The outcome appears on the welcome screen, next to the version number:

- **“Latest Version”** (green) — you are running the most recent version.
- **“Update Available v...”** (amber) — a newer version is available, shown by its number. It is a button: click it to open the update window.
- **“OFFLINE”** — the service could not be reached; try again later. The software works normally.

The check is optional and non-blocking: if you’re offline, RLMP starts and works without any trouble.

12.2 The update window

When an update is available, the dedicated window shows the current version, the new version and the **release notes**: the real list of what’s new in that version (the same changelog as this software), formatted and readable, not a bare list of files. The notes stay visible even once the download is complete, right before installing, so you always know what you’re about to apply. From here you decide:

- **Later** — closes the window without doing anything. You can reopen it whenever you like.
- **Download** — starts downloading the new version. The download **never starts on its own**: it begins only when you press this button. A progress bar shows its progress.
- **Restart and install** — appears when the download is complete: it closes the application and applies the update. If the open project has unsaved changes, RLMP asks what to do before installing, with the same choices as when closing: **Save** (save, then install), **Don’t Save** (install and discard the changes) or **Cancel** (install nothing; the app stays open). If saving fails, the installation doesn’t start. Once past this question, the shutdown is clean and immediate and the software doesn’t stay open behind the installer.

12.3 The “never during a live show” rule

The automatic check may find an update right while you’re on air. In that case, RLMP **does not interrupt you**: the update window waits and opens on its own only when the live show is over (when you stop everything). The priority is always the show in progress.

There is one exception, and it’s intentional: the **Check for updates now** button, in the Info panel (Tools menu), is an explicit action of yours and opens the window immediately, even live. If you press it, it’s because you want to.

12.4 Platform differences

How the update is installed depends on the operating system.

Windows and Linux (AppImage). The update is fully integrated: you download the new version from the window and the software installs it at the next restart, with no manual steps.

Linux (.deb package). With this format RLMP can't install the update reliably. Instead of automatic installation, the window notifies you and opens the browser on the download page for the new version: from there you download the package and install it as you would for a fresh install (Chapter 2). Your projects and `.lmp` files stay intact.

macOS (built from source). There is no official package to download: the window reports the new version and opens the project page, but to update you download the updated code and build again (section 2.3).

> **NOTE**
> In all cases, updating RLMP doesn't entail losing your projects: the `.lmp` files are compatible across versions and require no manual migration.

CHAPTER 13

Advanced features

This chapter gathers the features that sit outside the basic workflow but tend to stick once you find them: the NoteBoard, column colour management, transitions, the general settings, the launch log and the edit history.

13.1 NoteBoard: the script in the control room

The **NoteBoard** is the note system built into the clips. It lets you attach written text to any clip (operating instructions, running orders, notes on an interview, the full script of a spot) and have it appear automatically on screen the moment that clip starts playing.

Adding a note

1. Open the clip settings (right-click the card) and go to the Notes section.
2. Write the text in the free field. There is no length limit.

Clips with a note show a badge with a clipboard icon on the card.

The panel during the live show

When a clip with notes starts playing, the **NoteBoard panel** appears at the bottom of the screen with the associated text. The panel's header carries the words BROADCAST NOTES, an LED pulsing in the clip's colour and the clip's name. The panel stays visible for the whole duration of playback and closes on its own when the clip ends. If several clips with notes play together, the panel shows the one with the highest priority.

Use cases

- **Spoken production.** Attach to each ident the first lines of the spoken block that follows: when the ident starts, the text is already in front of your eyes.
- **Content to read.** An advertising spot with the full text in the note: as soon as it starts, you read it.
- **Operating instructions.** "Lower the monitor", "Check the guest's headphone level", "Start the recording".
- **Interviews.** The questions for the guest stay visible for the whole duration of the clip.

13.2 Column colour customization

The default colours have an established meaning (green for Assets, red for Songs, and so on), but every column is customizable. Click the button with the **coloured dot** in the column header: a palette of **30 colours** opens. Choose one and the column (header, cards, indicators) takes on the new colour immediately. The choice is saved in the project file.

The cards take on the column's colour: at rest only the thin strip at the top shows it, on air the whole card lights up. The text on a lit card turns dark or white depending on the colour, so it stays readable with any shade. Each project can thus have its own colour identity.

13.3 Transitions between clips

When a clip is set to Play Next, the passage to the next clip in the column happens according to the configured transition mode:

- **Crossfade.** The outgoing clip fades while the incoming one rises, overlapping. Default duration: 2 seconds.
- **Segue.** The outgoing clip fades out while the next one starts immediately at full volume. Default fade duration: 0.8 seconds.
- **Gapless (hard cut).** The outgoing clip stops abruptly and the next starts immediately, with no fade.

You can set a transition at the level of a single clip or leave **Global Default**, which applies the general choice defined in the Settings. The Pre-Show column uses crossfade as its default. All the modes can be tried out without going on air, using the "Test →" button in the editor (Chapter 5).

13.4 The General Settings window

The **Settings** (Tools menu) gather the software's global preferences, organized into tabs.

General

- **Language.** Select the interface language: English or Italian. The change is immediate.
- **Remote Control.** Turns on the browser remote for a phone, tablet or another computer. When on, it shows the QR code to scan, the PIN, the address and the connected devices, with the Copy link and New PIN buttons. The Turn on Remote Control when the app opens box, off by default, starts it automatically at every launch (Chapter 11).
- **Broadcast layout.** Shows or hides the grid columns individually. Hiding a column doesn't delete its clips: they stay in the project. It's a global preference, valid for all projects.

Audio & Mix

- **Output device.** The audio destination (Chapter 8).
- **Mixing intelligence.** The amount of ducking (how far the music drops when a voice speaks, default 20%) and its speed (default 500 ms).
- **Transitions.** The default transition mode and the crossfade and segue durations.

Microphone

- **Smart Mic — Auto-Ducking.** Enables the microphone that lowers the music when you speak (Chapter 6). At the top of the tab, a notice reminds you that the feature is designed for USB microphones connected directly to the computer, not for USB mixers.
- **Input Device.** The microphone to listen to: USB microphones and audio interfaces appear here automatically.
- **Activation threshold.** The level above which your voice triggers the ducking (default −30 dBFS); release happens 12 dB lower.
- **Activation and release hold.** How many milliseconds your voice must stay above the threshold before the music drops, and below it before the music comes back up.
- **Microphone Mix Channel.** With In Mix, your voice also goes into RLMP's output, with its own **Microphone Volume**. The **Bypass Master Chain** option decides how it comes out: off, the voice goes through a high-pass filter, compressor and limiter; on, it comes out as it is, with no added latency. A warning about the risk of **feedback** (howling), to be confirmed, reminds you to use headphones or a professional mixer: with the speakers on, your voice can leak back into the microphone.

Recording
A summary of the capture point (after the limiter) and the choice of default format proposed at export (Chapter 9).

Master Chain
- **Volume levelling.** Enables/disables loudness normalization and sets its target (default −16 LUFS).
- **Master Chain.** Enables or bypasses the whole chain, and adjusts the individual stages: HPF frequency, multiband glue style, limiter threshold. A button restores the defaults (Chapter 6).

13.5 Playout Log

The **Playout Log** (the **LOG** button in the left zone of the header) is the chronological launch log: it tracks what went on air and when, up to the last several thousand events. It's useful for reconstructing a running order after the fact, verifying what was broadcast, or compiling a report of the live show.

13.6 Undo and Redo

Changes to the running order (additions, moves, deletions) are reversible. `Ctrl+Z` undoes the last operation, `Ctrl+Y` (or `Ctrl+Shift+Z`) redoes it, with a history several dozen steps deep. The same entries are available in the Tools menu. It's the safety net for operations done in a hurry during preparation.

13.7 Toast notification system

RLMP doesn't use blocking windows for routine communications. Non-critical notifications appear as **toasts**: small, unobtrusive boxes at the bottom right of the screen that stay for a few seconds and disappear on their own without interrupting playback. They are used to confirm a save, the end of an export, a MIDI Learn operation, or to warn of missing files.

The **confirmation windows**, needed when an action is irreversible (deleting clips, closing an unsaved project), are instead modal and require a response, but they are designed not to cut off the playback in progress: the audio continues while you decide.

CHAPTER 14

Troubleshooting and FAQ

This chapter collects the most common problems in the daily use of Runtime Live Machine Pro, with their solutions. Each section describes the symptom, the most likely cause and the procedure to resolve it.

14.1 Audio problems

The timer runs and the VU meters move, but nothing is heard

The software is playing correctly (the signal is present on the internal bus), but it isn't reaching the monitoring device.

Check in order:

1. **Master Volume.** Is the slider in the header at zero? Bring it to 100%.
2. **Output device.** Open Settings → Audio & Mix and check which device is selected. Windows and macOS can change the identifier of USB devices when they are unplugged and reconnected. If the name doesn't match the one physically connected, select it again.
3. **External mixer.** If the signal reaches a hardware mixer, check that the channel fader isn't down or muted, and that the mixer output is connected to the monitors or the broadcast chain.

The audio skips, crackles or has dropouts

Under normal conditions the audio engine doesn't produce these artefacts. If they occur, the cause is almost always external to the software.

- **CPU under extreme load.** Close heavy applications running at the same time (video editing, rendering, intensive backups).
- **Audio buffer too low.** With a professional sound card, check the buffer value in the driver's control panel. A value of 256 or 512 samples is the right balance; below 128 samples, dropouts can appear.
- **Slow or stressed disk.** RLMP streams the audio from disk. A slow mechanical disk, or an almost-full SSD, can cause interruptions on large files.

The audio level is too low or too high

- **Per-clip Gain.** Adjust the Gain in the clip properties (right-click → Volume section).
- **Master Volume.** If the overall level is wrong, act on the slider in the header.
- **Levelling and Master Chain.** Volume levelling brings the clips' levels closer to a common reference; the Master Chain glue can make the sound more compact. If a result doesn't convince you, you can adjust or disable these stages in Settings → Master Chain.

14.2 Red clips and missing files

A card has turned red ("MISSING FILE") and doesn't respond to the click

The dark red card, with the MISSING FILE badge, indicates that the audio file isn't reachable at the path stored in the project.

Possible causes:

- The file was moved or renamed on disk.
- The file was on an external disk or a USB stick that is now disconnected.
- The project was opened on a different computer, where the paths don't match.

Solutions:

1. **Reconnect the disk.** If the file was on an external drive, plug it back in.
2. **Return the file to its original location.** If it was moved, put it back at the original path.
3. **Replace the clip.** Drag the correct file into the grid again and delete the red card.
4. **Use Export project with audio in future.** The most effective prevention is to consolidate the audio into the project before moving or transferring it (Chapter 10).

14.3 MIDI problems

The controller isn't detected

1. **Connection.** Check that the controller is connected and recognized by the operating system. RLMP detects the connection and disconnection of devices in real time; if it doesn't appear, unplug and reconnect the USB cable.
2. **Driver.** Most USB-MIDI controllers are class-compliant and need no driver. For professional surfaces with proprietary drivers, check that the driver is installed.
3. **Check in Learn mode.** Enable MIDI Learn and press a key on the controller: if the card receives the mapping, the controller is detected.

The mapped clips don't respond to the controller keys

- **MIDI Learn mode is still active.** In MIDI Learn the controller keys register new mappings instead of triggering the clips. Disable the mode from the Tools menu.
- **The mapping was lost.** The clip mappings are in the `.lmp` file; check that the project was saved after the MIDI Learn session. The global-function mappings are instead tied to the individual computer.

14.4 Startup problems

The application you built won't start on macOS (Gatekeeper warning)

See section 2.3: right-click the application → Open, or unblock it from System Settings → Privacy & Security.

The application won't start on Windows (SmartScreen warning)

See section 2.2. Click More info and then Run anyway.

Abnormal behaviour at startup

If the software behaves unexpectedly on opening, close and reopen RLMP. If the problem persists, check that the installation path doesn't contain special characters that might interfere with loading the FFmpeg components.

14.5 Frequently asked questions

Can RLMP run a radio station unattended for 24 hours? No. RLMP is designed for live production: shows attended by an operator. It has no hourly scheduling and no automatic playlist rotation. The Automix view offers a limited, voluntary automation of the music flow only, active while the view is open (Chapter 7). For 24/7 automation there are dedicated tools (Zara Radio, PlayIt Live, Rivendell): they answer different needs.

What's the difference between Save and Save As? Save Project overwrites the open `.lmp` file, silently. Save As… always opens the dialog box and creates a new file, without touching the current one.

Can I use RLMP on an iPad or on mobile devices? Not as the main application: RLMP is desktop software for Windows and Linux, and can also be built on macOS. A tablet or a phone can, however, act as a **remote** via the browser, through Remote Control (Chapter 11).

Are `.lmp` files from earlier versions compatible with 1.15.37? Yes. When you open a project created with an earlier version, RLMP automatically updates its structure, including the columns added in the meantime, without modifying the file until you perform a save.

How do I update RLMP to a new version? The software checks for updates at startup and notifies you. On Windows and Linux AppImage the installation is automatic from the update window; on Linux `.deb` the browser is opened on the download page; on macOS you build again from source. All the details are in Chapter 12.

Where are the automatic backups saved? In the `autosaves` folder inside the application data directory (`%APPDATA%\runtime-live-machine-pro\autosaves\` on Windows; equivalent paths on macOS and Linux, Chapter 10). The ten most recent snapshots are kept.

Does the software work offline? Yes, completely. RLMP doesn't require an internet connection to work. The network is used only for the update check (optional) and for Remote Control on the local network (optional).

Remote Control won't connect. Why? Check that Remote Control is on: by default it starts off at every launch, unless you have enabled automatic start in the Settings. Then check that the remote device is on the **same network** as the computer, and not on the guest network, which often isolates devices from each other. The easiest way to connect is to scan the **QR code**; if you type the address by hand, use the PIN shown in the Settings (it stays the same from one launch to the next, until you press New PIN). On Windows the program must be allowed through the firewall, on public networks too if your network is set as "Public". If the first address doesn't respond, try the others listed. If the switch turned itself off with a message, port 8787 is in use: close the other copy of RLMP or the program using it, then turn it back on. The details are in Chapter 11.

A pad tapped on the phone turned red with NOT RECEIVED. What does it mean? That the studio did not confirm the command within a second and a half: the effect did not start. Usually the Wi-Fi dropped for a moment or the phone was waking up. Look at the status line at the top of the page: when it shows **Connected** again, tap the pad again. Taps made without a connection are not sent later, on purpose.

In the MIX view a change was a crossfade instead of a beat-matched mix. Why? A beat-matched mix needs both tracks to have a reliably detected beat and tempos that aren't too different. The compatibility dot next to the track and the Last change line tell you the reason: BPM not detected, unreliable beat (typical of ballads and tracks without drums) or tempos too different. In these cases the software deliberately chooses a normal fade. Right after updating to 1.15.37 the tracks are analysed again once: until the analysis has finished, the changes are crossfades (Chapter 7).